MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE LDJ

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital AND-OR-INVERT gate module.

1.2 Module description. The following electrical parameters reflect the -55° to 100°C end-of-life limits from table II.

Name	Gate, AND-OR-INVERT
Family	Digital
Key code	LDJ
Size	1A
Weight	35 grams maximum
Failure rate Class I	0.22 failures/10^6 hours maximum
Class II	0.57 failures/10^6 hours maximum
Environment	Classes I and II as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	0.315 W (outputs not loaded)
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	-0.40 mA minimum, -1.73 mA maximum
High level (V_{IH} = 2.4 V)	0 µA minimum, 40 µA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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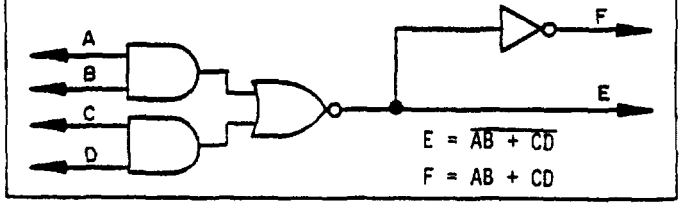
Output voltage	
Low level	
E and F outputs	0.0 V minimum, 0.4 V maximum
High level	
E and F outputs	2.4 V minimum, V_{CC} maximum
Output current	
Low level (V_{OL} = 0.4 V)	
E outputs	14.4 mA maximum
F outputs	16.0 mA maximum
High level (V_{OH} = 2.4 V)	
E outputs	360 µA minimum
F outputs	400 µA minimum
Propagation delay time	
t_{PLH}	
E outputs	4 ns minimum, 28 ns maximum
F outputs	10 ns minimum, 45 ns maximum
t_{PHL}	
E outputs	4 ns minimum, 22 ns maximum
F outputs	10 ns minimum, 43 ns maximum

1.3 Module function. This module contains six independent AND-OR-INVERT functions, with an additional inverter on each output as shown on figure 1.

1.3.1 Input to output relationship. Each AND-OR-INVERT gate has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship

Inputs				Outputs	
A	B	C	D	E	F
L	L	L	L	H	L
H	L	L	L	H	L
L	H	L	L	H	L
H	H	L	L	L	H
L	L	H	L	H	L
H	L	H	L	H	L
L	H	H	L	H	L
H	H	H	L	L	H
L	L	L	H	H	L
H	L	L	H	H	L
L	H	L	H	H	L
H	H	L	H	L	H
L	L	H	H	L	H
H	L	H	H	L	H
L	H	H	H	L	H
H	H	H	H	L	H



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1.3.2 Parallel outputs. Paralleling outputs is not permissible except when both inputs and outputs of circuits are paralleled to obtain extended drive. Each circuit whose output is paralleled with an output of another circuit must have each of its inputs paralleled with a corresponding input of the other circuit. Circuits may be paralleled only with other circuits within the same group. These circuit groups are:

(1) circuits 1 and 2
(2) Circuits 3 and 4
(3) Circuits 5 and 6

The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

1.3.3 Crosstalk relationships. The module has the following crosstalk relationships.

1.3.3 1 Module crosstalk. When connected in the worst case mode of operation (all outputs but one switching simultaneously), the LDJ module could exhibit substantial crosstalk. When a quiet F output is a low level, there may be as much as an 800 mV positive going spike. When a quiet E output is a low level, there may be as much as a 560 mV positive going spike

1.3.3.2 Circuit crosstalk The crosstalk on any output pin is nearly directly proportional to the number of switching outputs. Every switching F output can induce as much as 110 mV of noise on a low level quiet F output and as much as 50 mV of noise on a low level quiet E output. For every switching E output in Group A (circuits 1 and 2), the noise induced on a low level quiet E output in Group A is as much as 110 mV and the noise induced on a low level quiet F output in Group A is as much as 50 mV. For every switching E output outside Group A (circuits 3-6), the noise induced on a low level quiet E or F output is as much as 50 mV The preceding relationship applies to Group B (circuits 3 and 4) and Group C (circuits 5 and 6).

2. APPLICABLE DOCUMENTS

2 1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.22 for class I and 0.57 for class II failures per million operating hours when predicted in accordance with MIL-STD-1389.

3 4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design classes I and II as specified in MIL-STD-1389.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial and end-of-life limits		0°+5°C Initial and end-of-life limits		25° +5°C Initial and end-of-life limits		60°+5°C Initial and end-of-life limits		100°+5°C Initial and end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	0.01	0 8	0 01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B24	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C18	Static output voltage, low level	V	X	0.0	0.4	0.0	0 4	0.0	0.4	0.0	0.4	0.0	0.4	V
D1-D18	Static output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E24	Input current, low level	VII	X	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	mA
F1-F24	Input current, high level	VIII	X	0	40	0	40	0	30	0	40	0	40	µA
G1-G12	Output short circuit current	IX	X	20	55	20	55	20	55	20	55	20	55	mA
H1	Power supply current, low level	X	S	1	54	1	54	1	54	1	54	1	54	mA
H2	Power supply current, high level	X	S	1	57	1	57	1	57	1	57	1	57	mA
J1-J12	Transition time, t_{TLH}	XI	S	4	50	4	50	4	45	4	50	4	50	ns
K1-K12	Transition time, t_{THL}	XI	S	4	25	4	25	4	20	4	25	4	25	ns
L1,3,5, 7,9,11, 13,15,17, 19,21,23	Propagation delay time, t_{PHL} ("E" outputs)	XII	S	4	22	4	22	4	19	4	22	4	22	ns
L2,4,6, 8,10,12, 14,16,18, 20,22,24	Propagation delay time t_{PLH} ("F" outputs)	XII	S	10	45	10	45	10	43	10	45	10	45	ns
M1,3,5,7, 9,11,13, 15,17,19, 21,23	Propagation delay time, t_{PLH} ("E" outputs)	XIII	S	4	28	4	28	4	26	4	28	4	28	ns

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	-55° +5°C Initial and end-of-life limits		0°+5°C Initial and end-of-life limits		25° +5°C Initial and end-of-life limits		60°+5°C Initial and end-of-life limits		100°+5°C Initial and end-of-life limits		Unit
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M2,4.6 8, 10,12,14, 16,18,20, 22,24	Propagation delay time, t_{PHL} ("F" outputs)	XIII	S	10	43	10	43	10	41	10	43	10	43	ns
N1,3,5,7, 9,11	Output crosstalk, low level	XIV	Q	0	560	0	560	0	560	0	560	0	560	mV
N2,4,6,8, 10,12	Output crosstalk, low level	XIV	Q	0	800	0	800	0	800	0	800	0	800	mV
P1-P12	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
R	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient rise time	XVII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
T1	Isolation (initial) (end-of-life)	XVIII	S	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
T2,T3	Unused pin isolation (initial) (end-of-life)	XVIII	Q	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
U1-U24	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
V1-V24	Input clamp diode voltage	XX	Q	-0.4	-1.5	-0.4	-1 5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V

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TABLE II. Module electrical requirements -Continued.

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. Unless otherwise specified, limits are for initial and end-of-life.

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3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code The module key code shall be LDJ

3.5.2 Size. The module size shall be 1A

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40

3.5.5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/17-1 for class I and 17-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test All production modules shall be tested in accordance with notes 1 and 2 of table II

3 8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4.3 Test equipment The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	3
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Pulse generator	Item J01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Square wave generator	Item H01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	1
Megohm bridge	Item R06	1
Capacitance meter	Item C05	1

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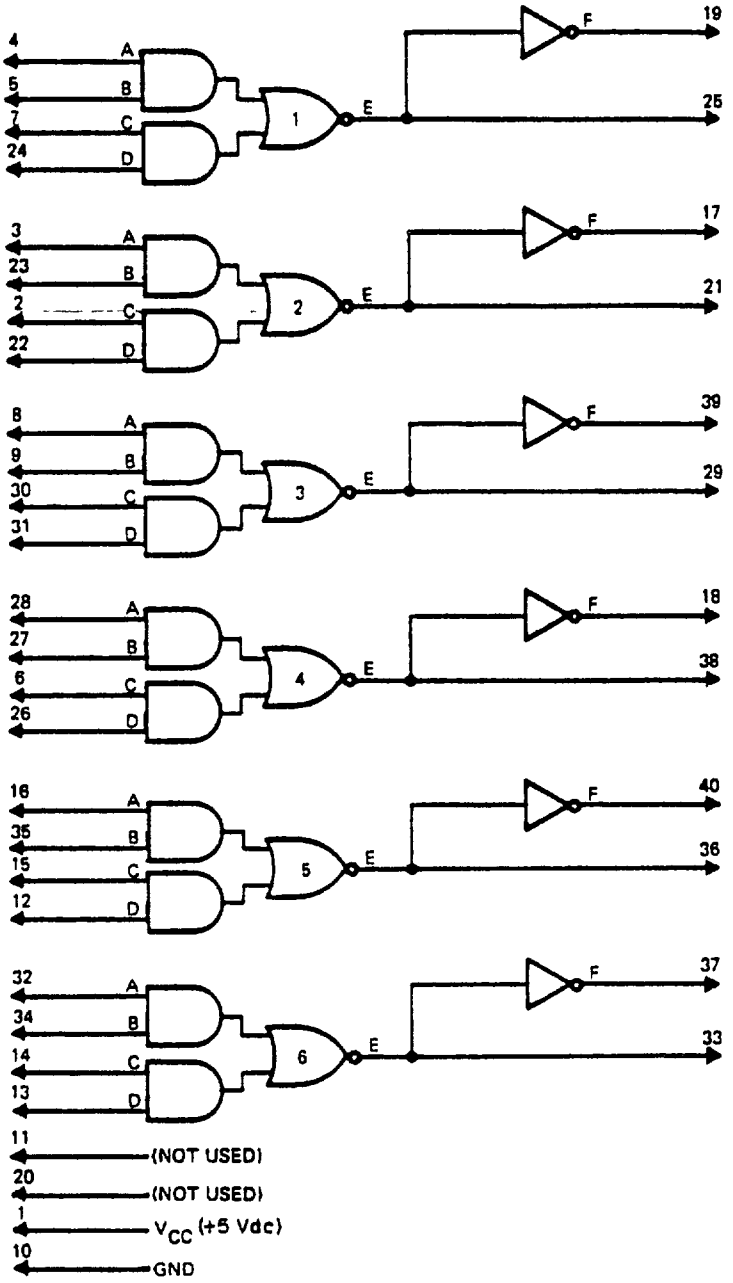


FIGURE 1. LDJ module function diagram.

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Resistors

453Ω ±1%, 1/4 W	6
316Ω ±1%, 1/4 W	6
357Ω ±1%, 1/4 W	1
51Ω ±5%, 1 W	2
6.04 kΩ ±1%, 1/4 W	6
Value as required for figure 8, note 1e, 1/4 W	6
6.65 kΩ ±1%, 1/4 W	6
Value as required for figure 8, note 2e, 1/4 W	6
511Ω ±1%, 1/4 W	1
887Ω ±1%, 1/4 W	1

Capacitors

100 pF ±20%	12
Value as required for figure 8, notes 1a and 2a	12
Value as required for figure 8, notes 1b and 2b	12

Diode

1N4150 or equivalent	12

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4 4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B or T are part of a sequence of tests, the order of testing shall be as follows. A, B, T where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4 5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through V in table II shall be performed in the order specified in 4 4.2 using the initial limits at 25° ±5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P,S, and V, shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test conditions S, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through S and V shall be performed in the order specified in 4 4.2 using the operating temperature end-of-life requirements of table II Following the return of the module to 25°C, electrical tests C through S,U, and V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II

5. PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

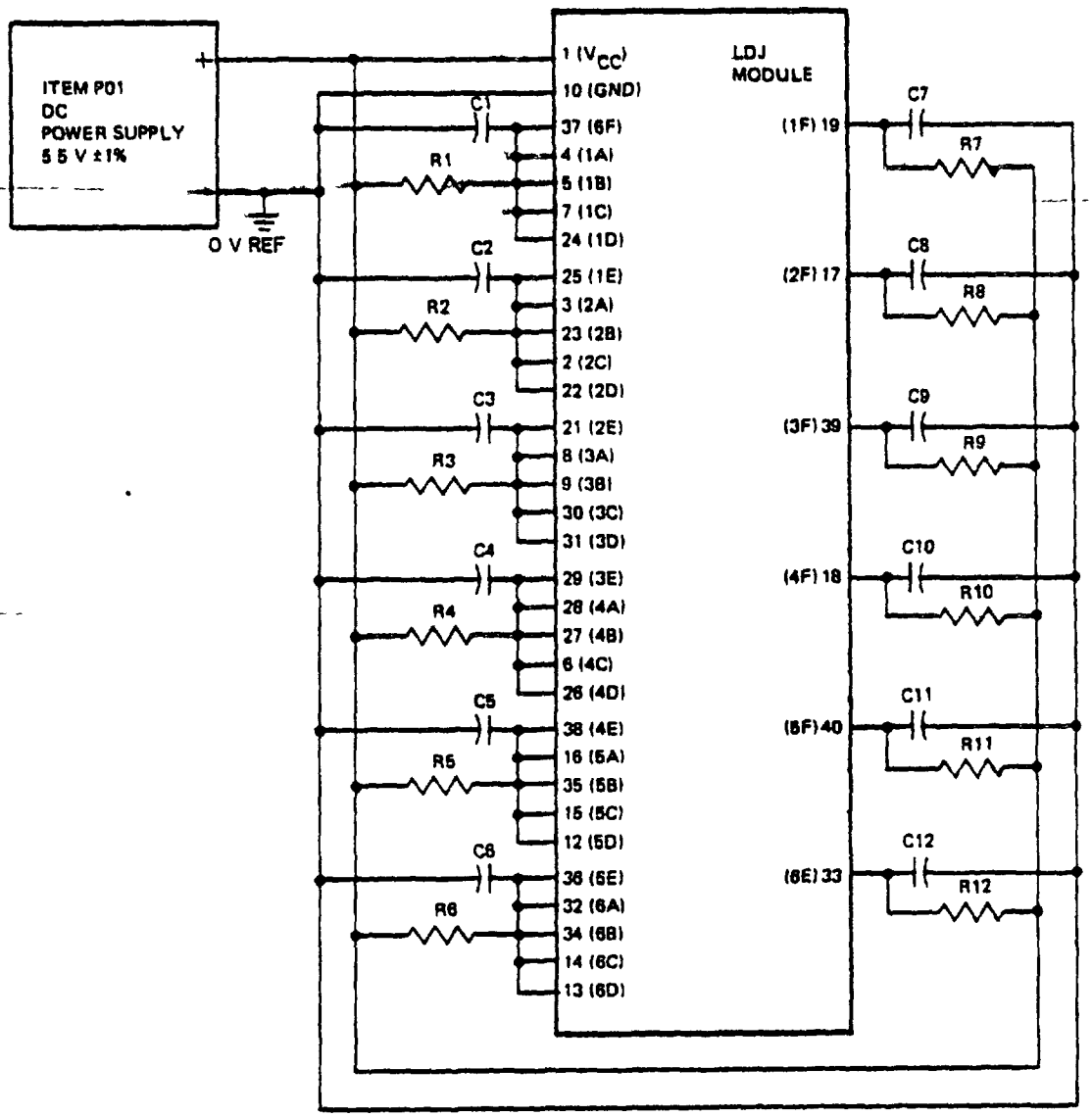
6.3 Cross reference of classification. Part number M28787/17-1 supersedes part number M28787/17. The two parts are interchangeable.

Preparing activity:
Navy - EC

Agent.
DLA - ES

(Project 5963-N020)

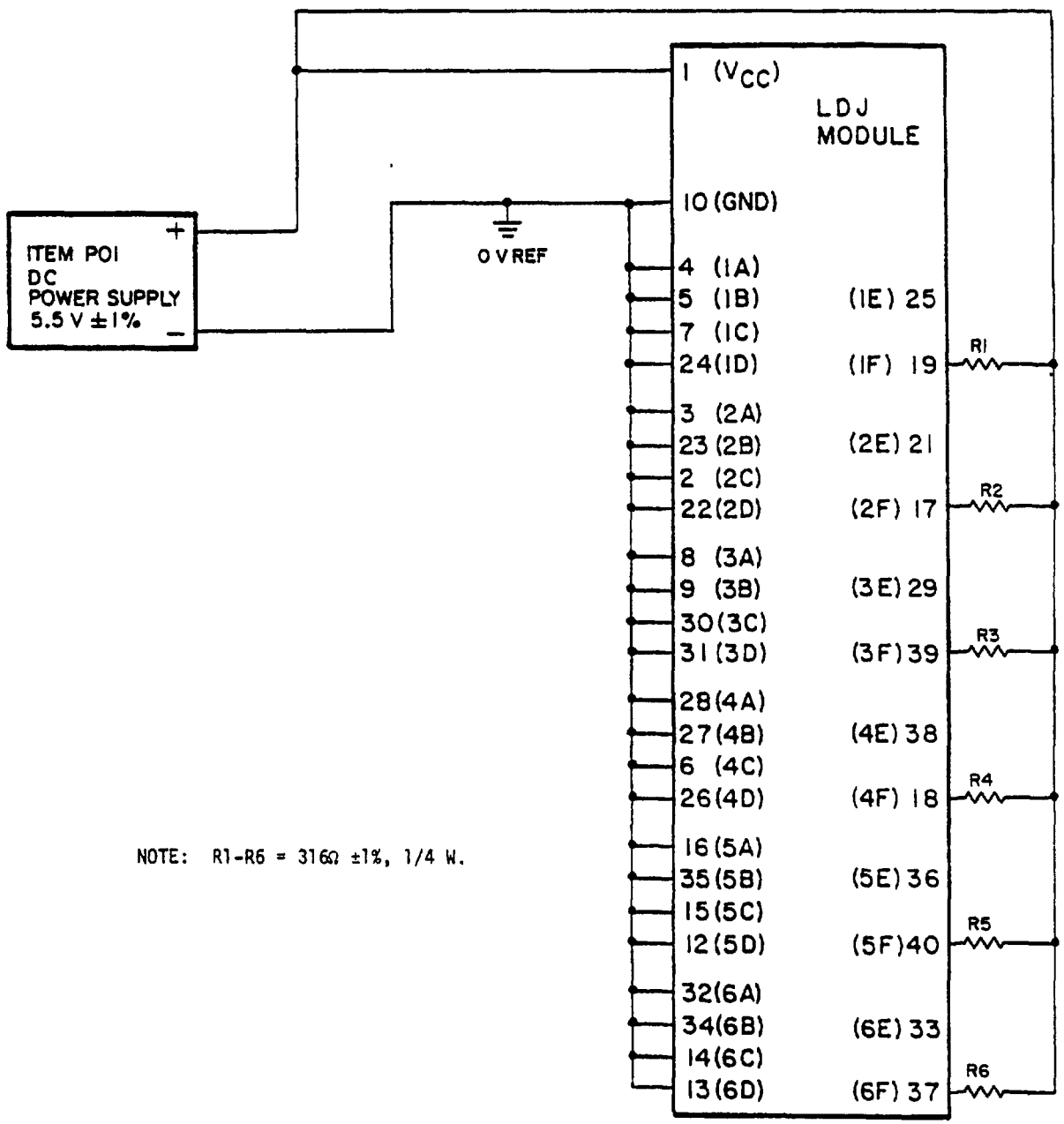
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NOTES:
1. R1-R6 = 453Ω ±1%, 1/4 W.
2. R7-R12 = 316Ω ±1%, 1/4 W.
3. C1-C12 = 100 pF ±20%.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R6 = 316Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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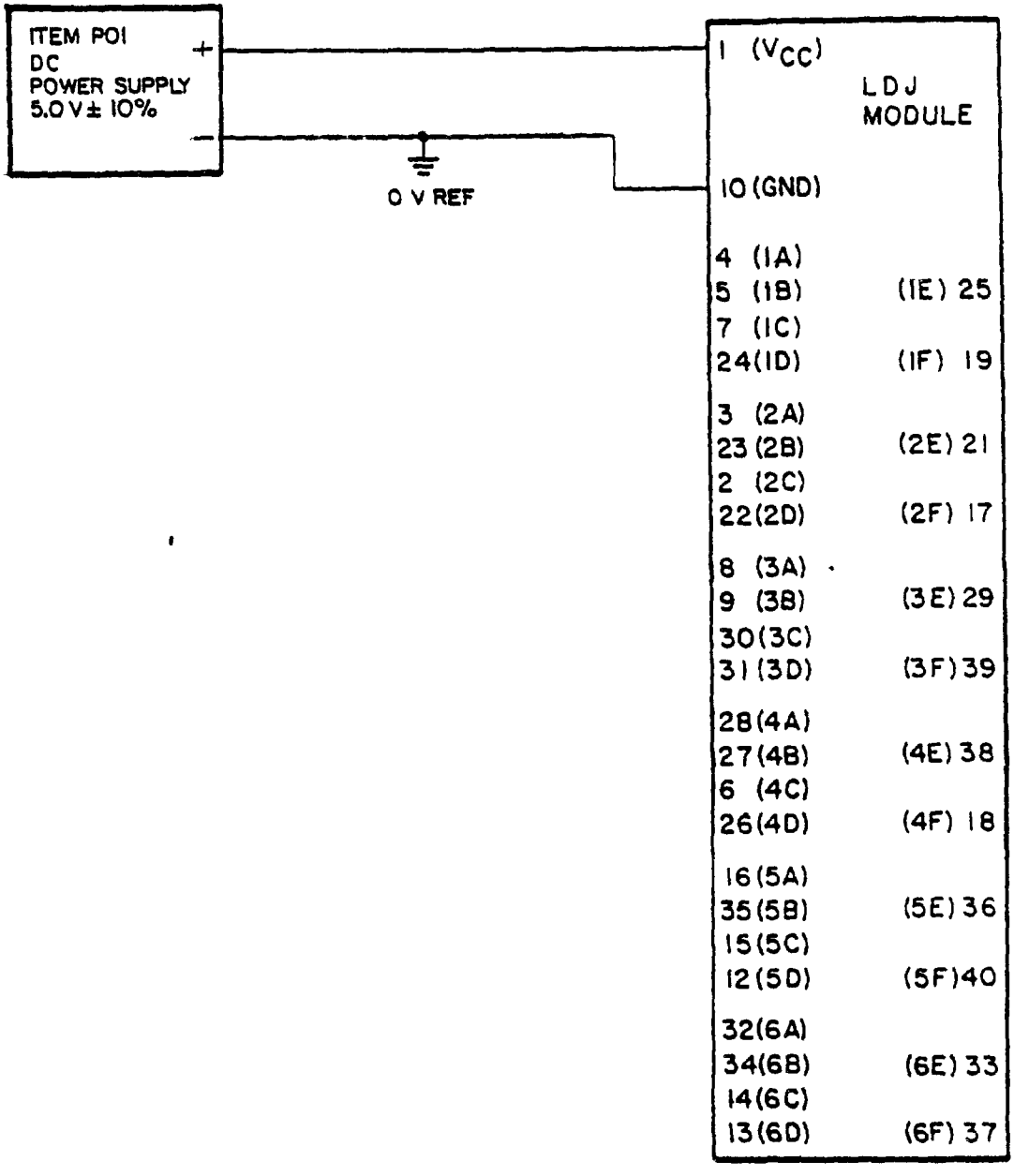


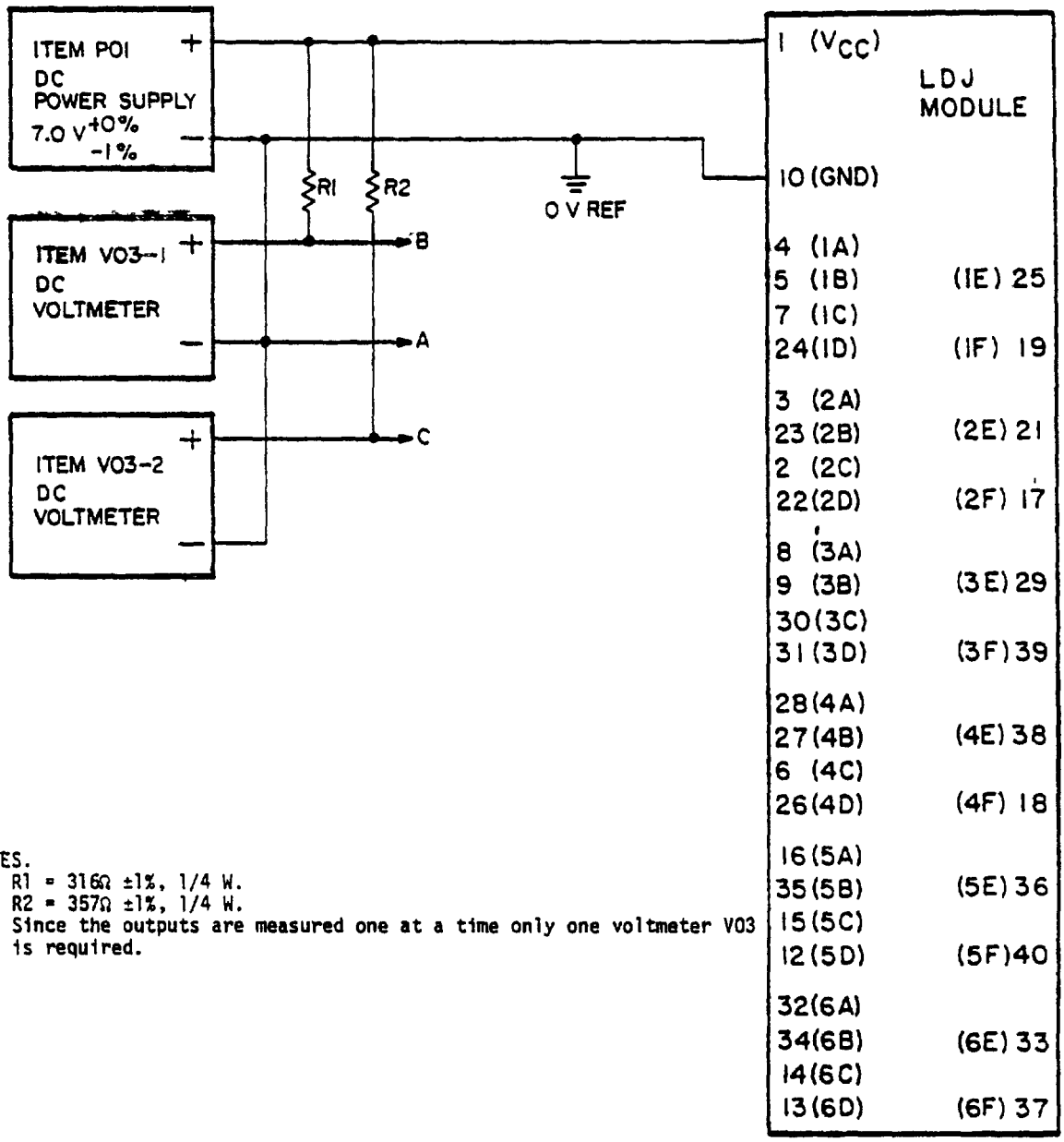
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
A1	1E V_{OUT} (low-loaded)	Open	Open	25	0.01	0.8	V	5
A2	2E			21				
A3	3E			29				
A4	4E			38				
A5	5E			36				
A6	6E	Open	Open	33				
A7	1F	5,7	19	Open				
A8	2F	2,3	17					
A9	3F	8,30	39					
A10	4F	26,27	18					
A11	5F	15,16	40					
A12	6F V_{OUT} (low-loaded)	14,34	37	Open	0.01	0.8	V	5

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NOTES.
1. R1 = 316Ω ±1%, 1/4 W.
2. R2 = 357Ω ±1%, 1/4 W.
3. Since the outputs are measured one at a time only one voltmeter V03 is required.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III

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TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	4	5	0.0	1.0	mA	6
B2	1B	5	4				
B3	1C	7	24				
B4	1D	24	7				
B5	2A	3	23				
B6	2B	23	3				
B7	2C	2	22				
B8	2D	22	2				
B9	3A	8	9				
B10	3B	9	8				
B11	3C	30	31				
B12	3D	31	30				
B13	4B	27	28				
B14	4A	28	27				
B15	4C	6	26				
B16	4D	26	6				
B17	5A	16	35				
B18	5B	35	16				
B19	5D Input breakdown current	12	15	0.0	1.0	mA	6

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TABLE IV. Input breakdown current test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B20	5C Input breakdown current	15	12	0.0	1.0	mA	6
B21	6A	32	34				
B22	6B	34	32				
B23	6D	13	14				
B24	6C Input breakdown current	14	13	0.0	1.0	mA	6

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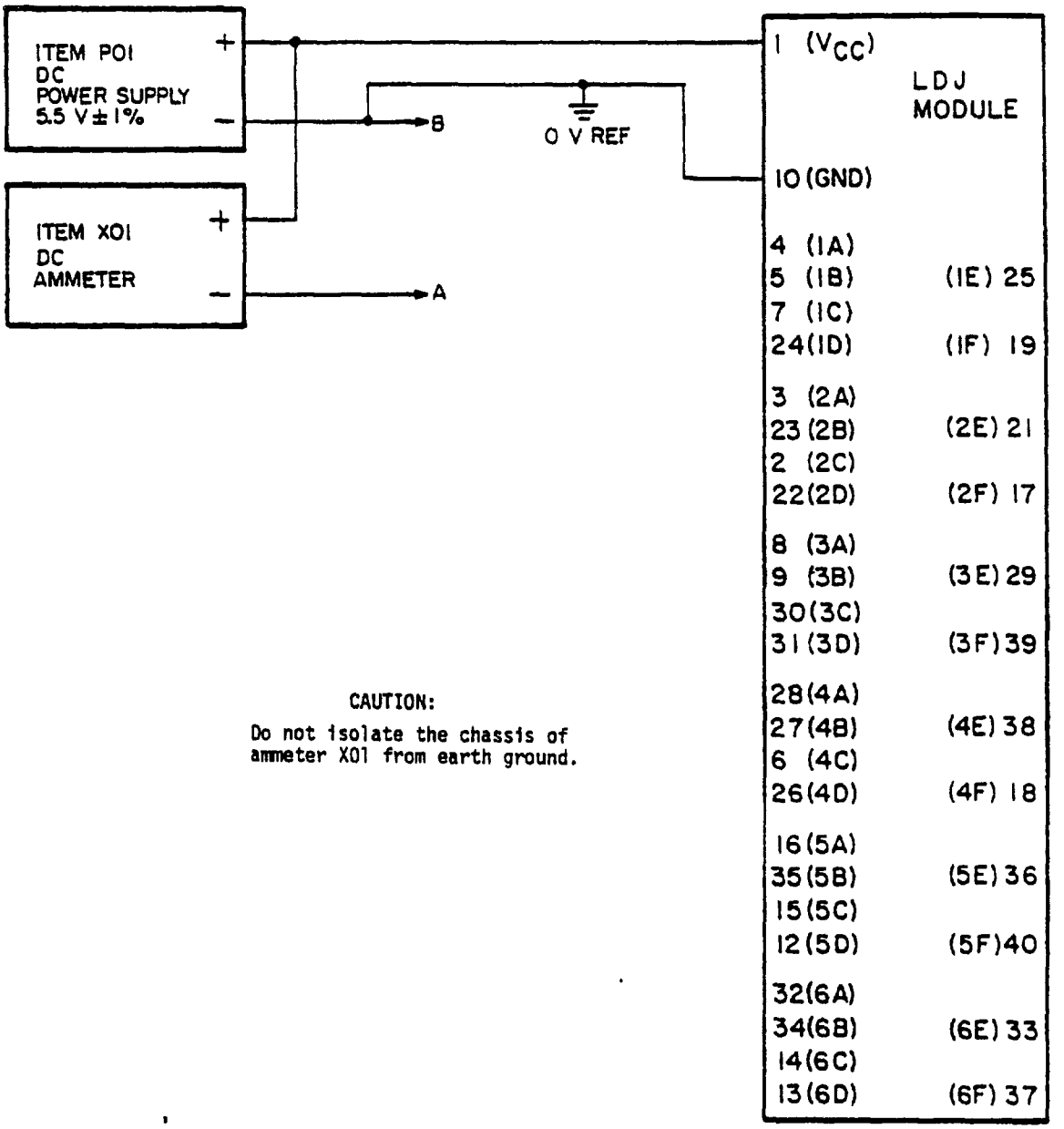


FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Static output voltage (low-level) test.

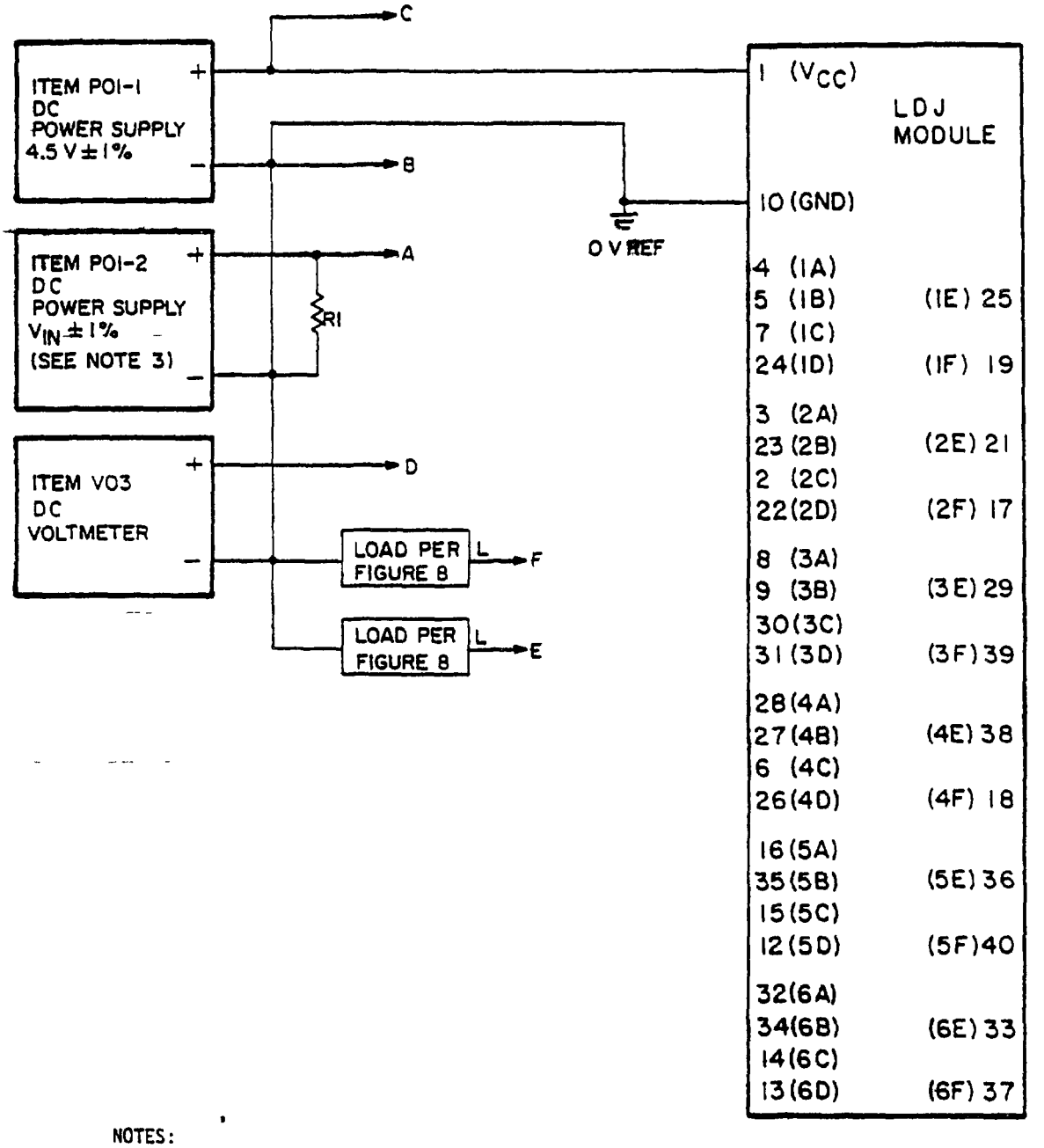
Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
C1	Static output voltage, low level	4,5	7,24	Open	25	25	19	0.0	0.4	V	7
C2		7,24	4,5		25	25	19				
C3		3,23	2,22		21	21	17				
C4		2,22	3,23		21	21	17				
C5		8,9	30,31		29	29	39				
C6		30,31	8,9		29	29	39				
C7		27,28	6,26		38	38	18				
C8		6,26	27,28		38	38	18				
C9		16,35	12,15		36	36	40				
C10		12,15	16,35		36	36	40				
C11		32,34	13,14		33	33	37				
C12		13,14	32,34	Open	33	33	37				
C13		4,7	Open	5,24	19	25	19				
C14		2,3		22,23	17	21	17				
C15		8,30		9,31	39	29	39				
C16		6,27		26,28	18	38	18				
C17		12,16		15,35	40	36	40				
C18	Static output voltage, low level	13,32	Open	14,34	37	33	37	0.0	0.4	V	7

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TABLE VI. Static output voltage (high-level) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
D1	Static output voltage, high level	4,7	Open	5,24	25	25	19	2.4	4.5	V	7
D2		5,24		4,7	25	25	19				
D3		2,3		22,23	21	21	17				
D4		22,23		2,3	21	21	17				
D5		8,30		9,31	29	29	39				
D6		9,31		8,30	29	29	39				
D7		6,27		26,28	38	38	18				
D8		26,28		6,27	38	38	18				
D9		12,16		15,35	36	36	40				
D10		15,35		12,16	36	36	40				
D11		13,32		14,34	33	33	37				
D12		14,34	Open	13,32	33	33	37				
D13		4,5	7,24	Open	19	25	19				
D14		3,23	2,22		17	21	17				
D15		8,9	30,31		39	29	39				
D16		27,28	6,26		18	38	18				
D17		16,35	12,15		40	36	40				
D18	Static output voltage, high level	32,34	13,14	Open	37	33	37	2.4	4.5	V	7

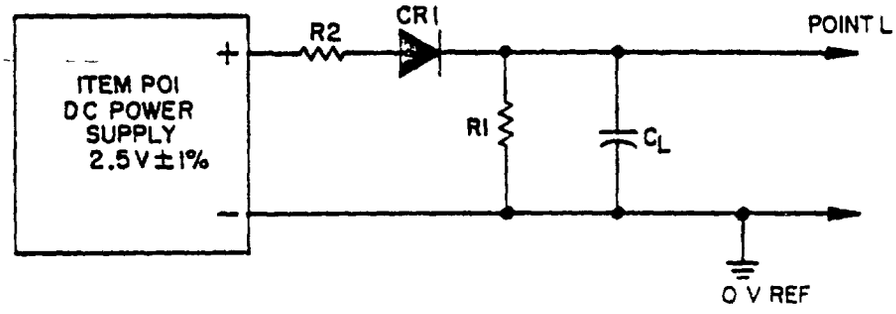
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. V_{IN} = 2.0 V for test conditions C1-C12 and D13-D18. V_{IN} = 0.8 V for test conditions C13-C18 and D1-D12.

FIGURE 7. Static output voltage (low-level and high-level) test circuit for tables V and VI.

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NOTES:
1. Load circuit 1 for outputs 1F-6F:
   a. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions J, K, L and M.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N and P.
   c. CR1 = 1N4150 or equivalent.
   d. R1 = 6.04 kΩ ±1%, 1/4 W.
   e. R2 = Select value for current flow of 17.3 mA ±1% from point L to ground.
   f. C_L is not required for dc electrical tests C1-C18 and D1-D18.
   g. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
   h. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

2. Load circuit 2 for outputs 1E-6E:
   a. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions J, K, L and M.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N and P.
   c. CR1 = 1N4150 or equivalent.
   d. R1 = 6.65 kΩ ±1%, 1/4 W.
   e. R2 = Select value for current flow of 16.6 mA ±1% from point L to ground.
   f. C_L is not required for dc electrical tests C1-C18 and D1-D18.
   g. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
   h. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 8. Logic load circuit.

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TABLE VII. Input current (low-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1A	Input current, low level	5	4	Open	-0.40	-1.73	mA	9
E2	1B		4	5					
E3	1C		24	7					
E4	1D		7	24					
E5	2A		23	3					
E6	2B		3	23					
E7	2C		22	2					
E8	2D		2	22					
E9	3A		9	8					
E10	3B		8	9					
E11	3C		31	30					
E12	3D		30	31					
E13	4B		28	27					
E14	4A		27	28					
E15	4C		26	6					
E16	4D		6	26					
E17	5A		35	16					
E18	5B		16	35					
E19	5C	Input current, low level	12	15	Open	-0.40	-1.73	mA	9

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TABLE VII. Input current (low-level) test -Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E20	5D Input current, low level	15	12	Open	-0.40	-1.73	mA	9
E21	6A	34	32					
E22	6B	32	34					
E23	6D	14	13					
E24	6C Input current, low level	13	14	Open	-0.40	-1.73	mA	9

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TABLE VIII. Input current (high-level) test.

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1A	Input current, high level	4	Open	5	0	30	µA	9
F2	1B		5		4				
F3	1C		7		24				
F4	1D		24		7				
F5	2A		3		23				
F6	2B		23		3				
F7	2C		2		22				
F8	2D		22		2				
F9	3A		8		9				
F10	3B		9		8				
F11	3C		30		31				
F12	3D		31		30				
F13	4B		27		28				
F14	4A		28		27				
F15	4C		6		26				
F16	4D		26		6				
F17	5A		16		35				
F18	5B		35		16				
F19	5C	Input current, high level	12	Open	15	0	30	µA	9

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TABLE VIII. Input current (high-level) test -Continued.

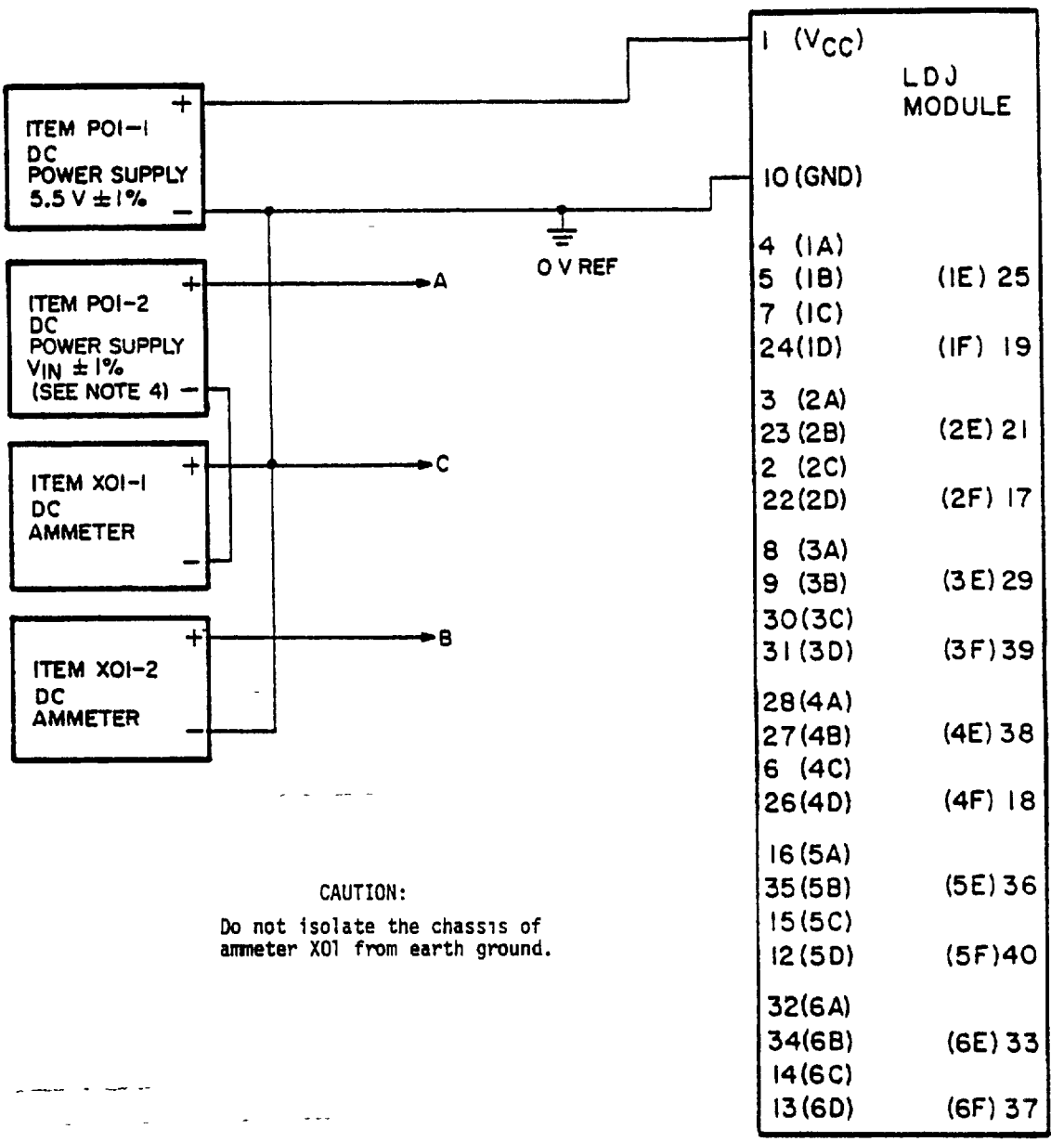
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F20	5D Input current, high level	15	Open	12	0	30	µA	9
F21	6A	32		34				
F22	6B	34		32				
F23	6D	13		14				
F24	6C Input current, high level	14	Open	13	0	30	µA	9

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TABLE IX. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1E Output short circuit current	Open	25	4,7	20	55	mA	9
G2	2E		21	2,3				
G3	3E		29	8,30				
G4	4E		38	6,27				
G5	5E		36	15,16				
G6	6E	Open	33	14,34				
G7	1F	4,5	19	Open				
G8	2F	3,23	17					
G9	3F	8,30	39					
G10	4F	26,27	18					
G11	5F	15,16	40					
G12	6F Output short circuit current	14,34	37	Open	20	55	mA	9

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since the ammeters are not used simultanesouly, only 1 ammeter XOI is required.
2. With point "A" disconnected, ammeter XOI should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only 1 output shall be shorted at a time and measurement taken within 5 seconds.
4. V_{IN} = 4.5 V for test conditions E1-E24 and G7-G12, V_{IN} = 2.4 V for test conditions F1-F24, and open for test conditions G1-G6.

FIGURE 9. Input current (low-level and high-level) test and output short circuit current test circuit for tables VII, VIII and IX.

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TABLE X. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Power supply current, low-level	Open	Open	1	54	mA	10
H2	Power supply current, high level	B	A	1	57	mA	10

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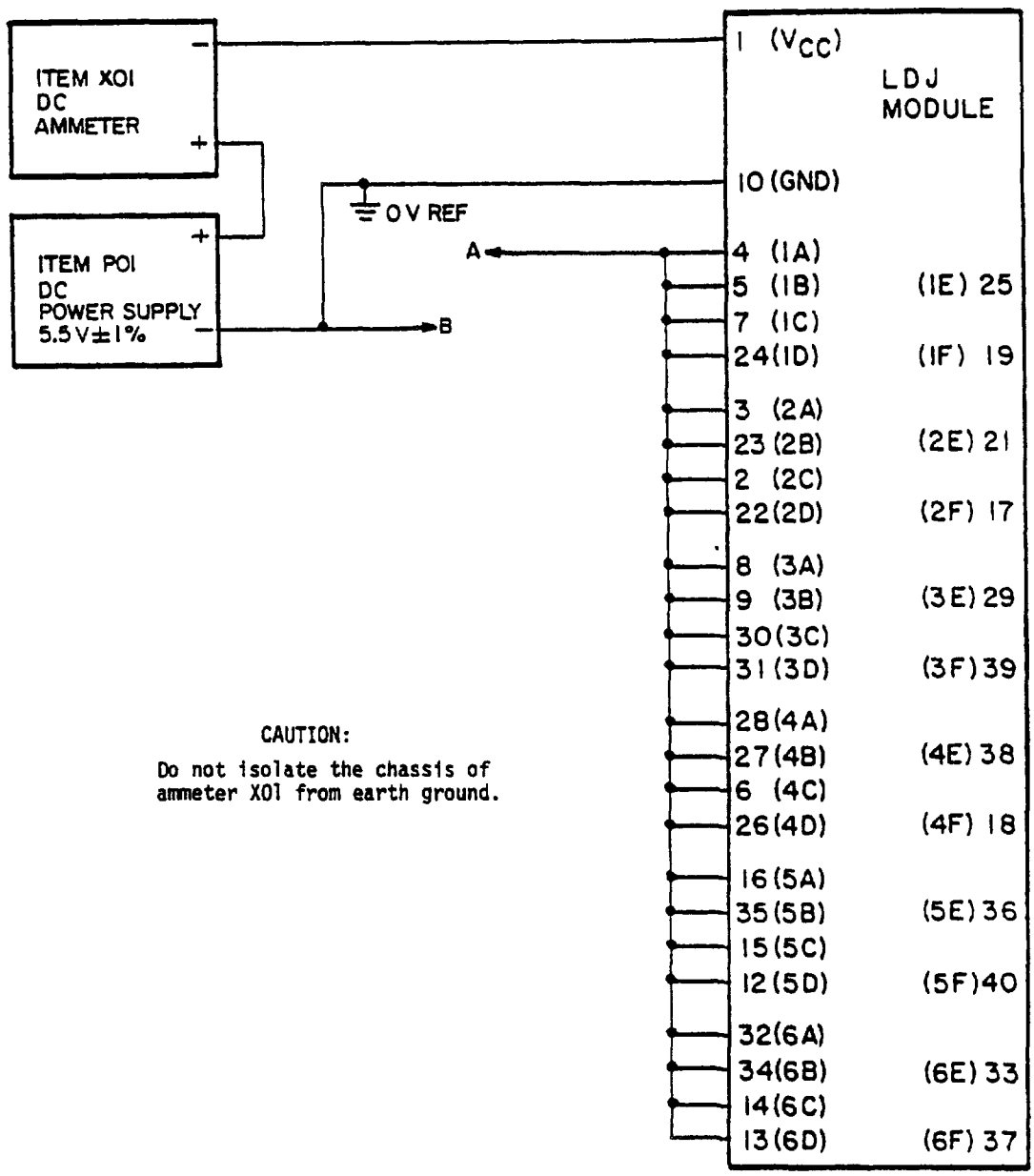


FIGURE 10. Power supply current (low-level and high-level) test circuit for table X.

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TABLE XI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Initial limits 25° ±5°C ambient t_{TLH} Min	t_{TLH} Max	Initial limits 25° ±5°C ambient t_{THL} Min	t_{THL} Max	Unit	Test figure
J1, K1	1E Transition time	4	5	7	25	25	19	4	45	4	20	ns	11
J2, K2	1F	4	5	7	19	25	19						
J3, K3	2E	3	23	2	21	21	17						
J4, K4	2F	3	23	2	17	21	17						
J5, K5	3E	8	9	30	29	29	39						
J6, K6	3F	8	9	30	39	29	39						
J7, K7	4E	27	28	6	38	38	18						
J8, K8	4F	27	28	6	18	38	18						
J9, K9	5E	16	35	15	36	36	40						
J10, K10	5F	16	35	15	40	36	40						
J11, K11	6E	32	34	13	33	33	37						
J12, K12	6F Transition time	32	34	13	37	33	37	4	45	4	20	ns	11

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L1	Propagation delay time, t_{PHL} ("E" output)	4	5	7,24	25	25	19	4	19	ns	11
L3	↓	7	24	4,5	25	25	19	↓	↓	↓	↓
L5		3	23	2,22	21	21	17				
L7		2	22	3,23	21	21	17				
L9		8	9	30,31	29	29	39				
L11		30	31	8,9	29	29	39				
L13		27	28	6,26	38	38	18				
L15		6	26	27,28	38	38	18				
L17		16	35	12,15	36	36	40				
L19		12	15	16,35	36	36	40				
L21		32	34	13,14	33	33	37				
L23	Propagation delay time, t_{PHL} ("E" output)	13	14	32,34	33	33	37	4	19		
L2	Propagation delay time, t_{PLH} ("F" output)	4	5	7,24	19	25	19	10	43		
L4	↓	7	24	4,5	19	25	19	↓	↓		
L6		3	23	2,22	17	21	17				
L8		2	22	3,23	17	21	17				
L10	Propagation delay time, t_{PLH} ("F" output)	8	9	30,31	39	29	39	10	43	ns	11

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test -Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
L12	Propagation delay time, t_{PLH} ("F" output)	30	31	8,9	39	29	39	10	43	ns	11
L14		27	28	6,26	18	38	18				
L16		6	26	27,28	18	38	18				
L18		16	35	12,15	40	36	40				
L20		12	15	16,35	40	36	40				
L22		32	34	13,14	37	33	37				
L24	Propagation delay time, t_{PLH} ("F" output)	13	14	32,34	37	33	37	10	43	ns	11

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

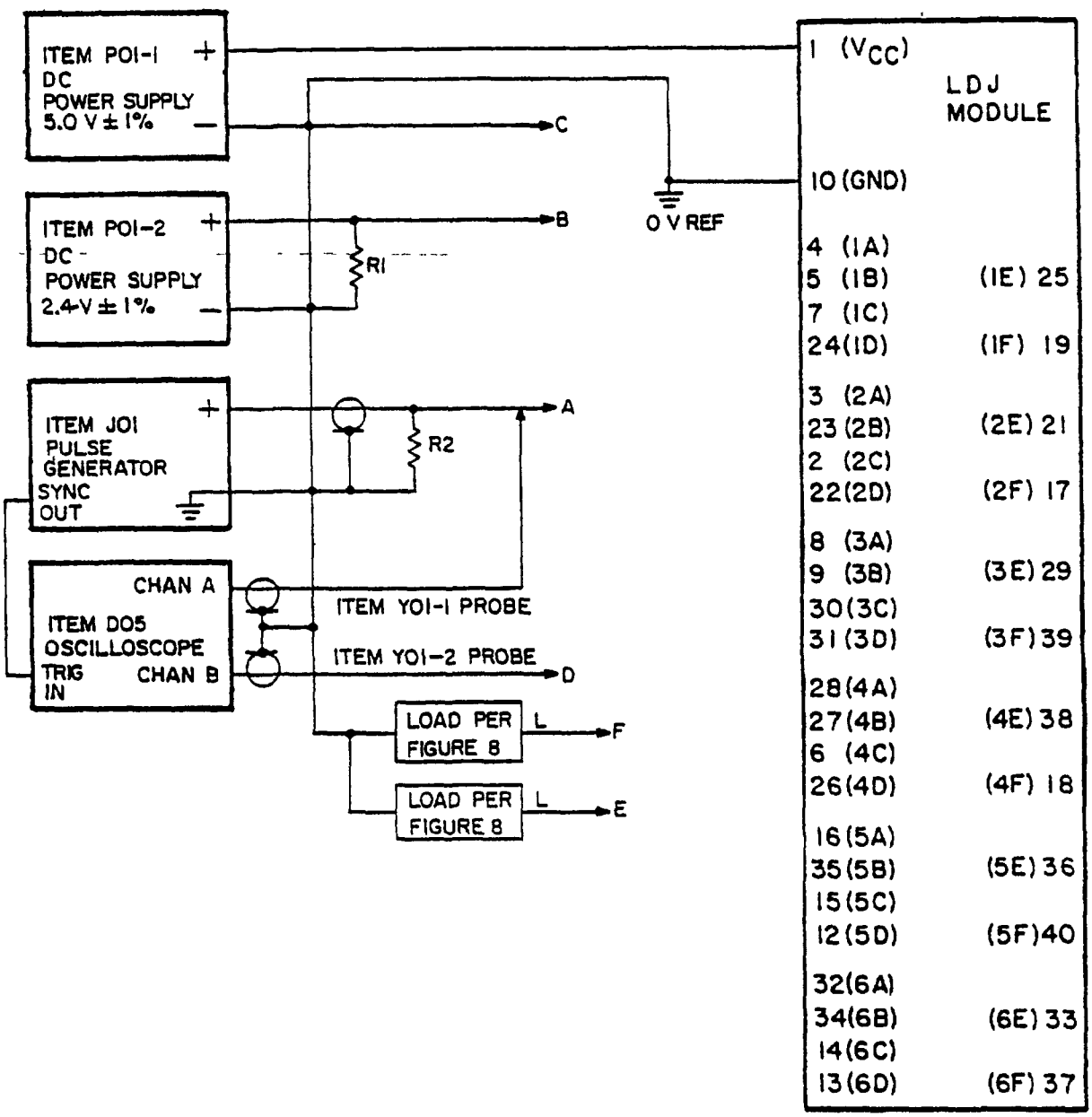
Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M1	Propagation delay time, t_{PLH} ("E" output)	4	5	7,24	25	25	19	4	26	ns	11
M3		7	24	4,5	25	25	19				
M5		3	23	2,22	21	21	17				
M7		2	22	3,23	21	21	17				
M9		8	9	30,31	29	29	39				
M11		30	31	8,9	29	29	39				
M13		27	28	6,26	38	38	18				
M15		6	26	27,28	38	38	18				
M17		16	35	12,15	36	36	40				
M19		12	15	16,35	36	36	40				
M21		32	34	13,14	33	33	37				
M23	Propagation delay time, t_{PLH} ("E" output)	13	14	32,34	33	33	37	4	26		
M2	Propagation delay time, t_{PHL} ("F" output)	4	5	7,24	19	25	19	10	41		
M4		7	24	4,5	19	25	19				
M6		3	23	2,22	17	21	17				
M8		2	22	3,23	17	21	17				
M10	Propagation delay time, t_{PHL} ("F" output)	8	9	30,31	39	29	39	10	41	ns	11

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test -Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
M12	Propagation delay time, t_{PHL} ("F" output)	30	31	8,9	39	29	39	10	41	ns	11
M14		27	28	6,26	18	38	18				
M16		6	26	27,28	18	38	18				
M18		16	35	12,15	40	36	40				
M20		12	15	16,35	40	36	40				
M22		32	34	13,14	37	33	37				
M24	Propagation delay time, t_{PHL} ("F" output)	13	14	32,34	37	33	37	10	41	ns	11

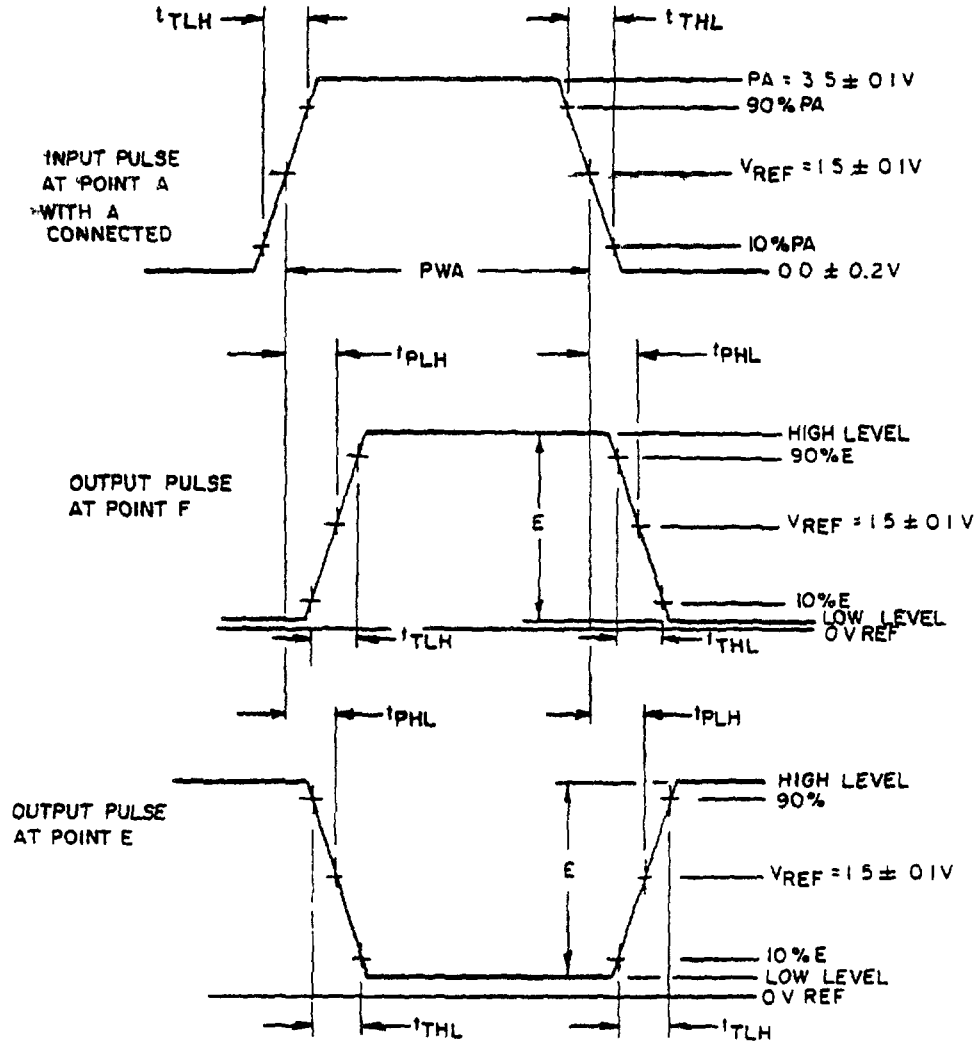
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Both outputs of the circuit under test shall be loaded simultaneously.

FIGURE 11. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit and waveforms for tables XI, XII and XIII.

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NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PRR = 5 MHz ±10%.
2. All voltage levels are referenced to ground.
3 Before making the measurements, verify that the output high level $\geq$ 2.4 V and the low level $\leq$ 0.4 V

FIGURE 11 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit and waveforms for tables XI, XII and XIII. - Continued.

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TABLE XIV. Output crosstalk, (low-level) test.

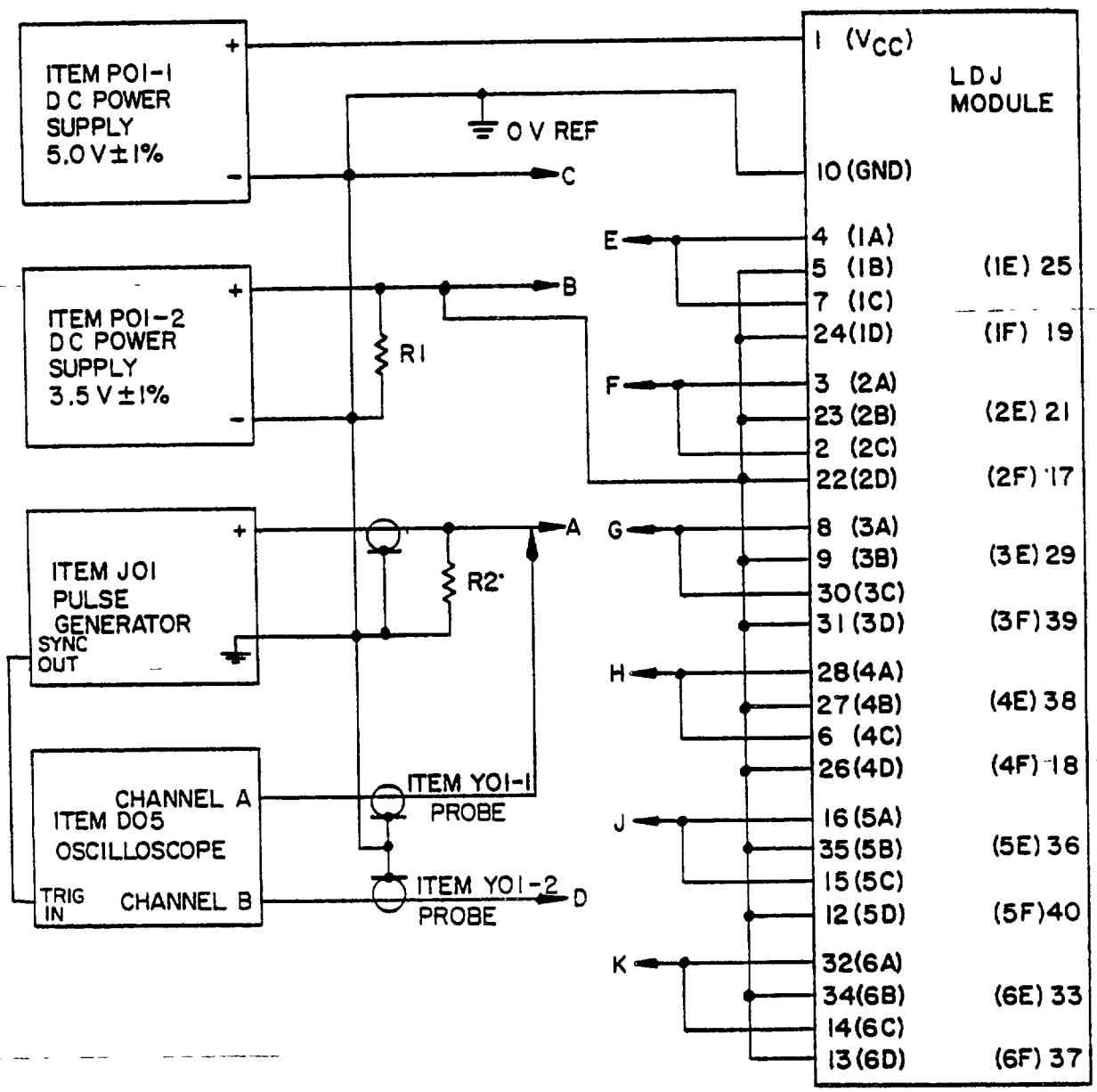
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1E Output crosstalk, low level	F,G,H, J,K	E	Open	25	0	560	mV	12
N2	1F	F,G,H, J,K	Open	E	19		800		
N3	2E	E,G,H, J,K	F	Open	21		560		
N4	2F	E,G,H, J,K	Open	F	17		800		
N5	3E	E,F,H, J,K	G	Open	29		560		
N6	3F	E,F,H, J,K	Open	G	39		800		
N7	4E	E,F,G, J,K	H	Open	38		560		
N8	4F	E,F,G, J,K	Open	H	18		800		
N9	5E	E,F,G, H,K	J	Open	36		560		
N10	5F	E,F,G, H,K	Open	J	40		800		
N11	6E	E,F,G, H,J	K	Open	33		560		
N12	6F Output crosstalk, low level	E,F,G, H,J	Open	K	37	0	800	mV	12

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TABLE XV. Output crosstalk,(high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1E Output crosstalk, high level	F,G,H, J,K	Open	E	25	0	400	mV	12
P2	1F	F,G,H, J,K	E	Open	19				
P3	2E	E,G,H, J,K	Open	F	21				
P4	2F	E,G,H, J,K	F	Open	17				
P5	3E	E,F,H, J,K	Open	G	29				
P6	3F	E,F,H, J,K	G	Open	39				
P7	4E	E,F,G, J,K	Open	H	38				
P8	4F	E,F,G, J,K	H	Open	18				
P9	5E	E,F,G, H,K	Open	J	36				
P10	5F	E,F,G, H,K	J	Open	40				
P11	6E	E,F,G, H,J	Open	K	33				
P12	6F Output crosstalk, high level	E,F,G, H,J	K	Open	37	0	400	mV	12

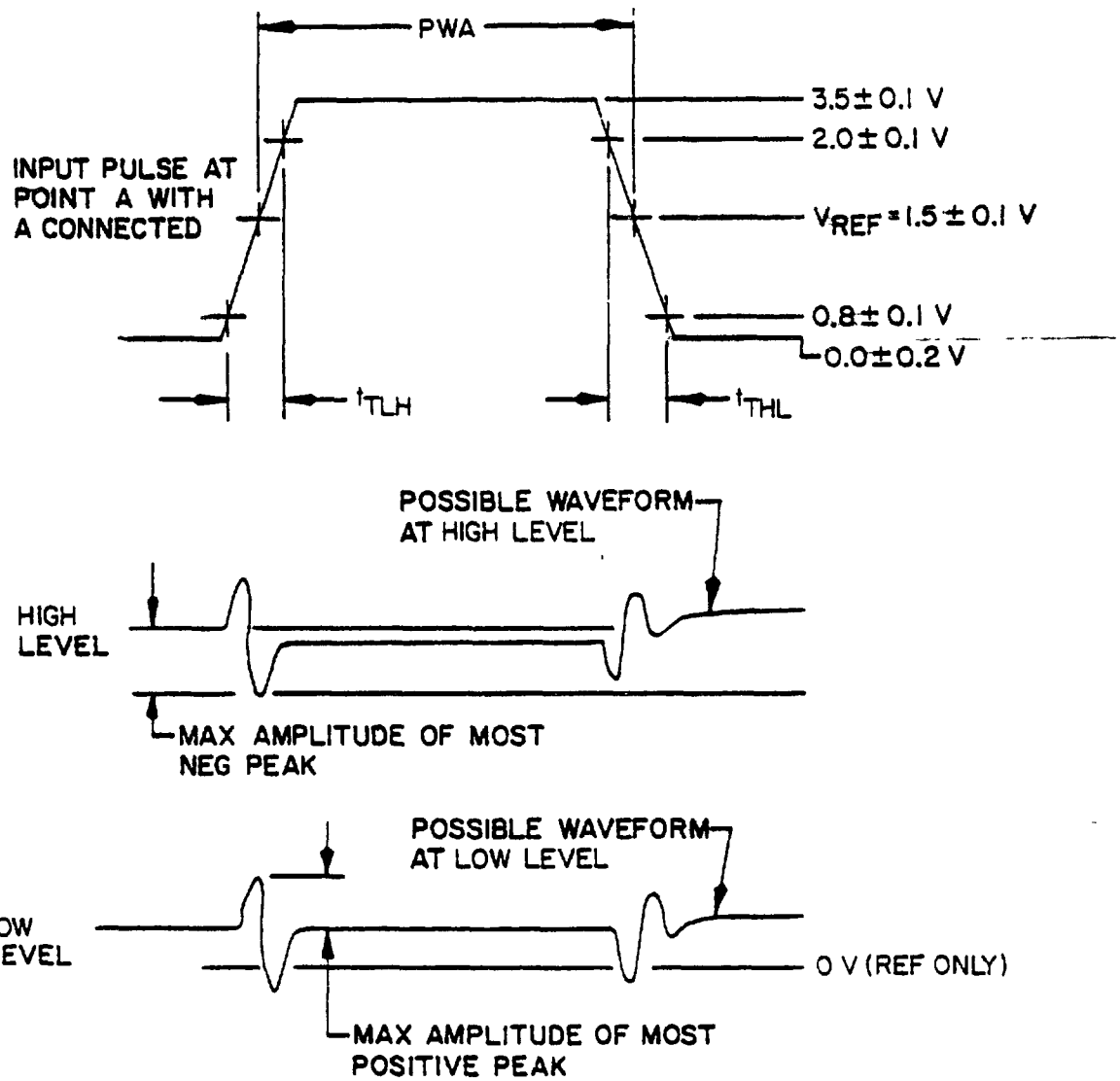
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load all outputs with the loads as specified on figure 8.

FIGURE 12. Output crosstalk (low-level and high-level) test circuit and waveforms for tables XIV and XV.

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NOTE: Input pulse parameters:
PWA = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3 5 ±1.5 ns (verified at each input pin)
PRR = 1 MHz ±10%.

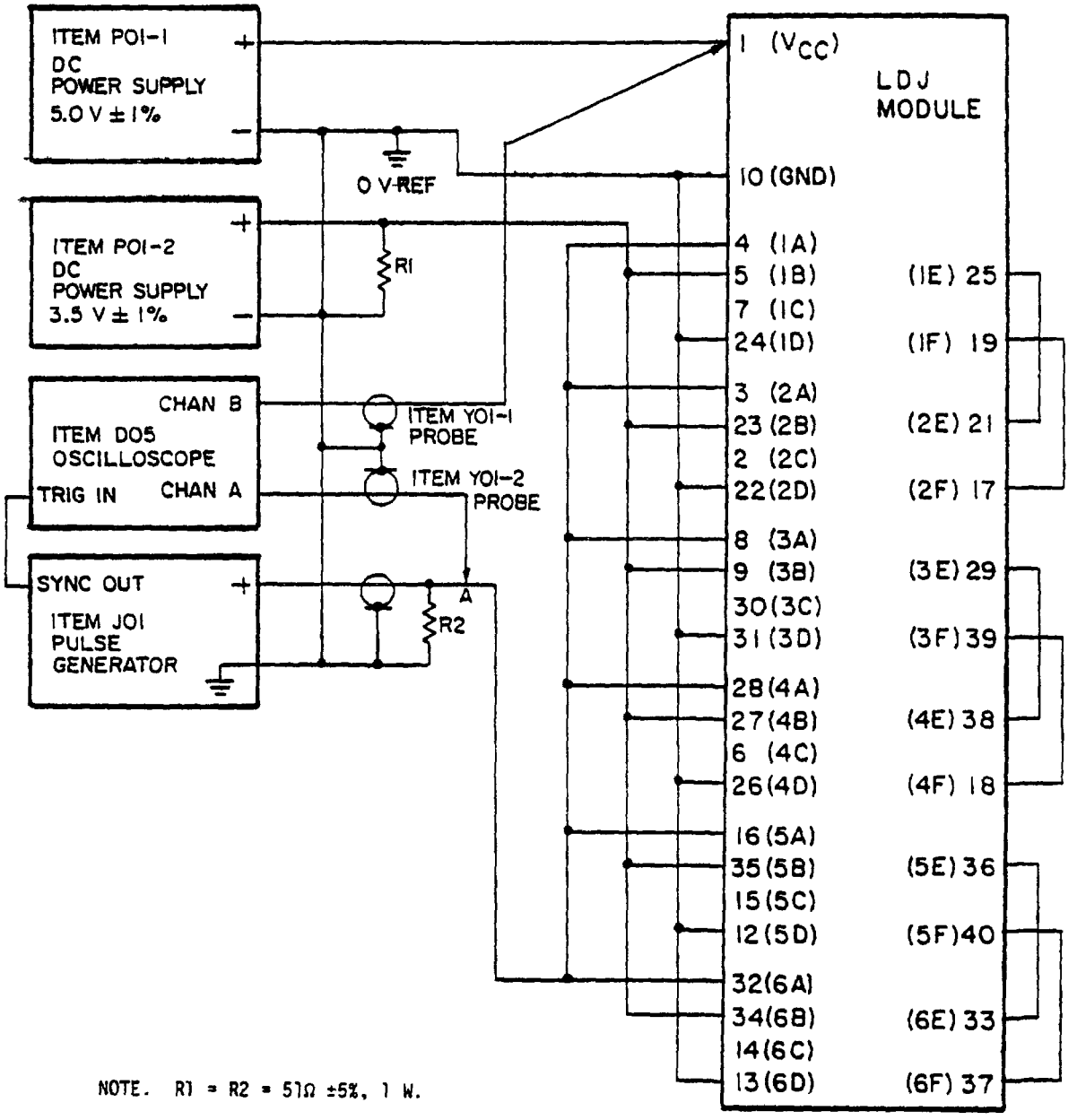
FIGURE 12. Output crosstalk (t_{TLH} and t_{THL}) test circuit and waveforms for tables XIV and XV - Continued.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	13

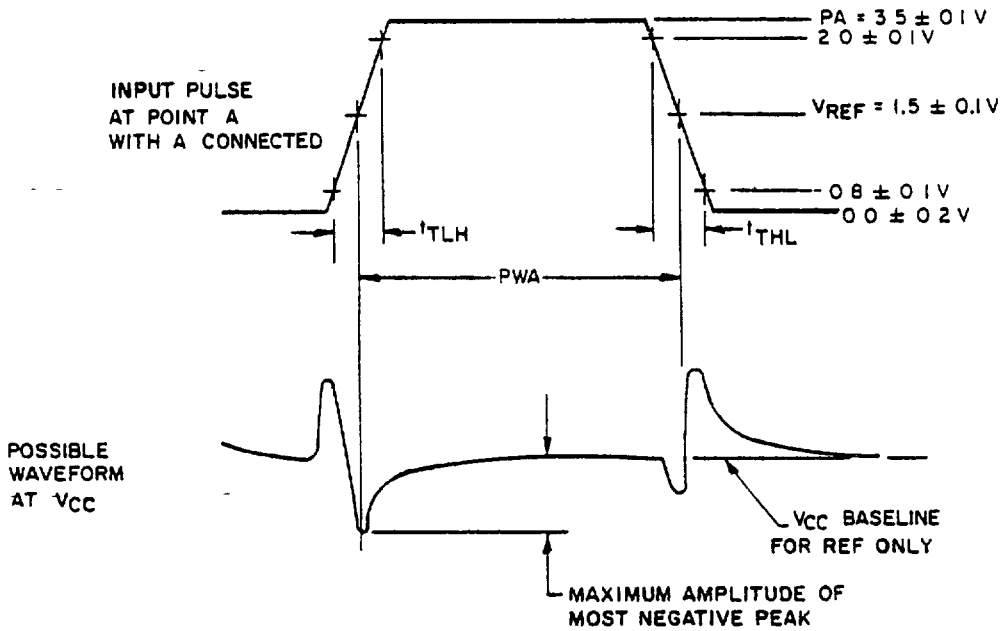
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NOTE. R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 13. Power supply transient amplitude test circuit and waveforms for table XVI.

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NOTE Input pulse parameters
PWA = 3 µs ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin)
PRR = 100 kHz ±10%.

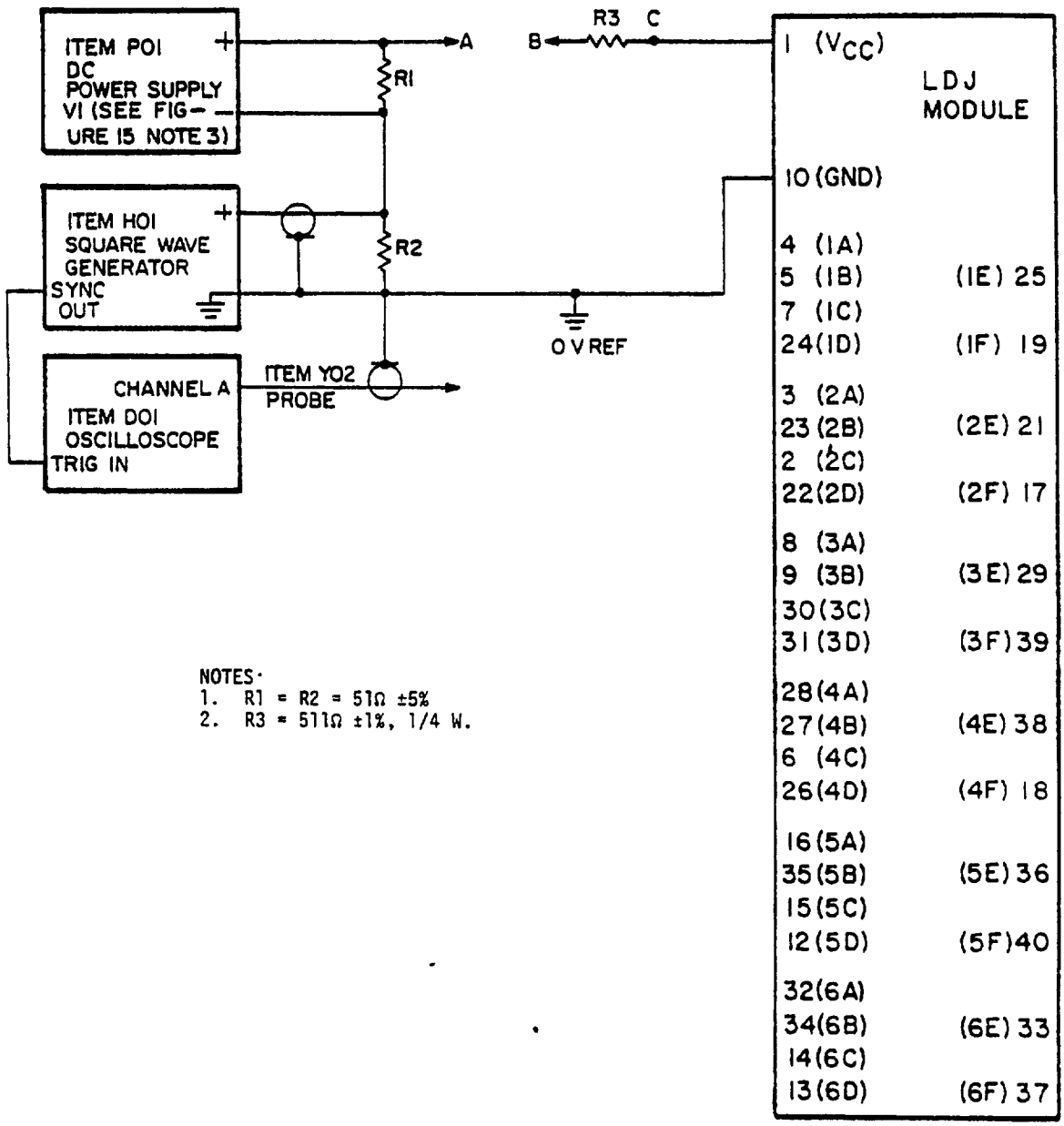
FIGURE 13. Power supply transient amplitude test circuit and waveforms for table XVI - Continued.

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TABLE XVII. Power supply transient (rise time) test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
S	Power supply transient, rise time	1.0	2.0	ms	14,15

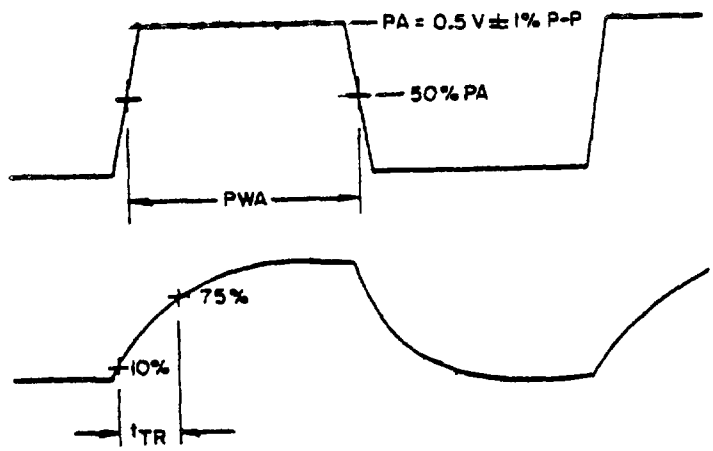
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NOTES:
1. R1 = R2 = 51Ω ±5%
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 14. Power supply transient rise time test circuit for table XVII.

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INPUT PULSE AT POINT A

DOWN LEVEL

WAVEFORM AT POINT C

NOTES:
1. Input pulse parameters:
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   Rise and fall time of input < 100 ns (reference).
2. With point "A" open, adjust input as shown.
3. With point "A" open, adjust V1 such that the "down" level of the input is at 0 ±0.01 V.
4. Connect "A" to "B"; recheck the "down" level and pulse parameters at point "A".
5. Measure transient rise time (t_{TR}) at point "C".

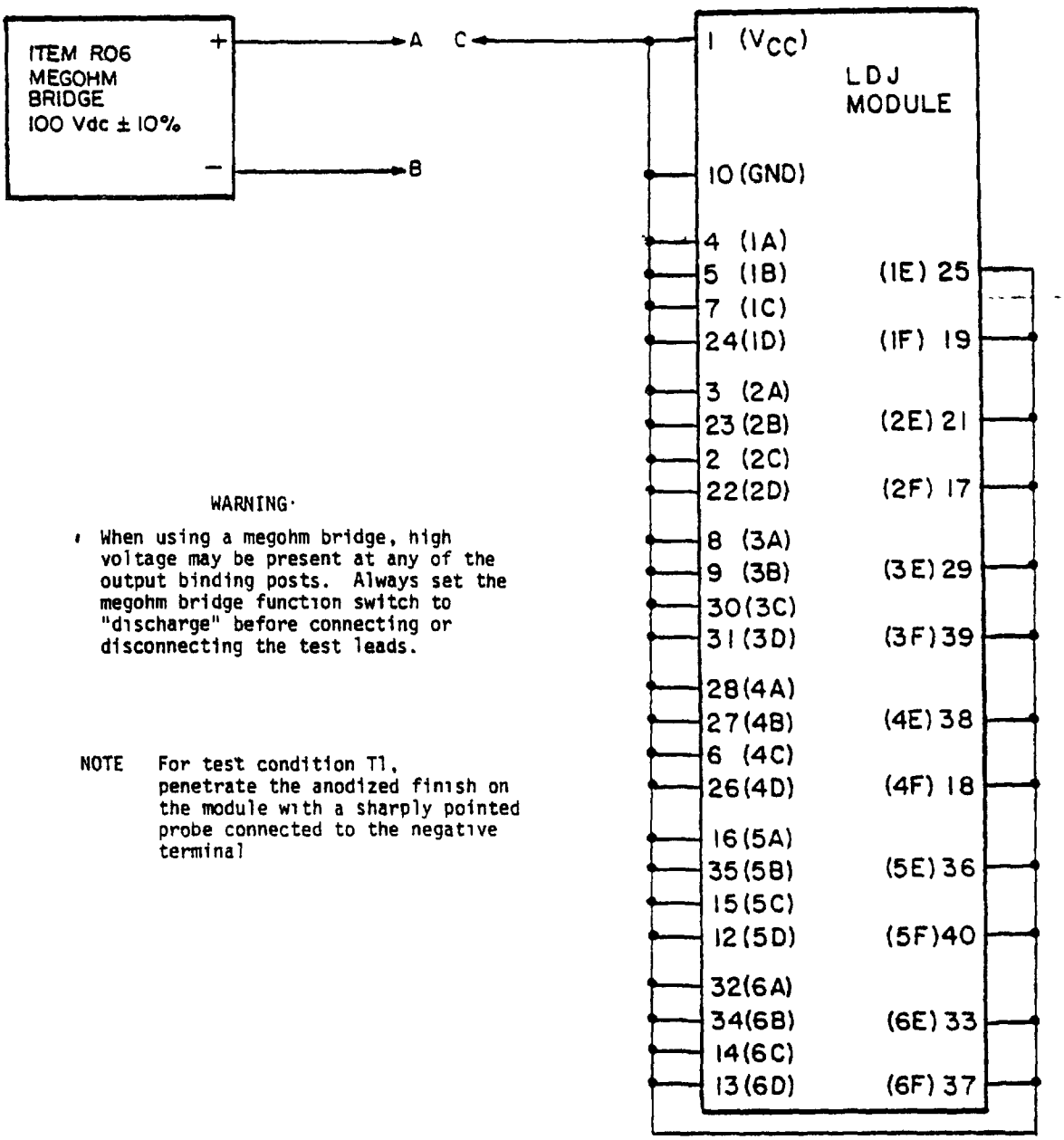
FIGURE 15. Power supply transient rise time test circuit waveforms for table XVII.

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TABLE XVIII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Isolation	C, 11, 20	Frame	10	∞	MΩ	16
T2	Isolation, unused pins	C	11,20	10	∞	MΩ	16
T3	Isolation, unused pins	11	20	10	∞	MΩ	16

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WARNING:

- When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE For test condition T1, penetrate the anodized finish on the module with a sharply pointed probe connected to the negative terminal

FIGURE 16. Isolation test circuit for table XVIII.

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TABLE XIX. Input capacitance test.

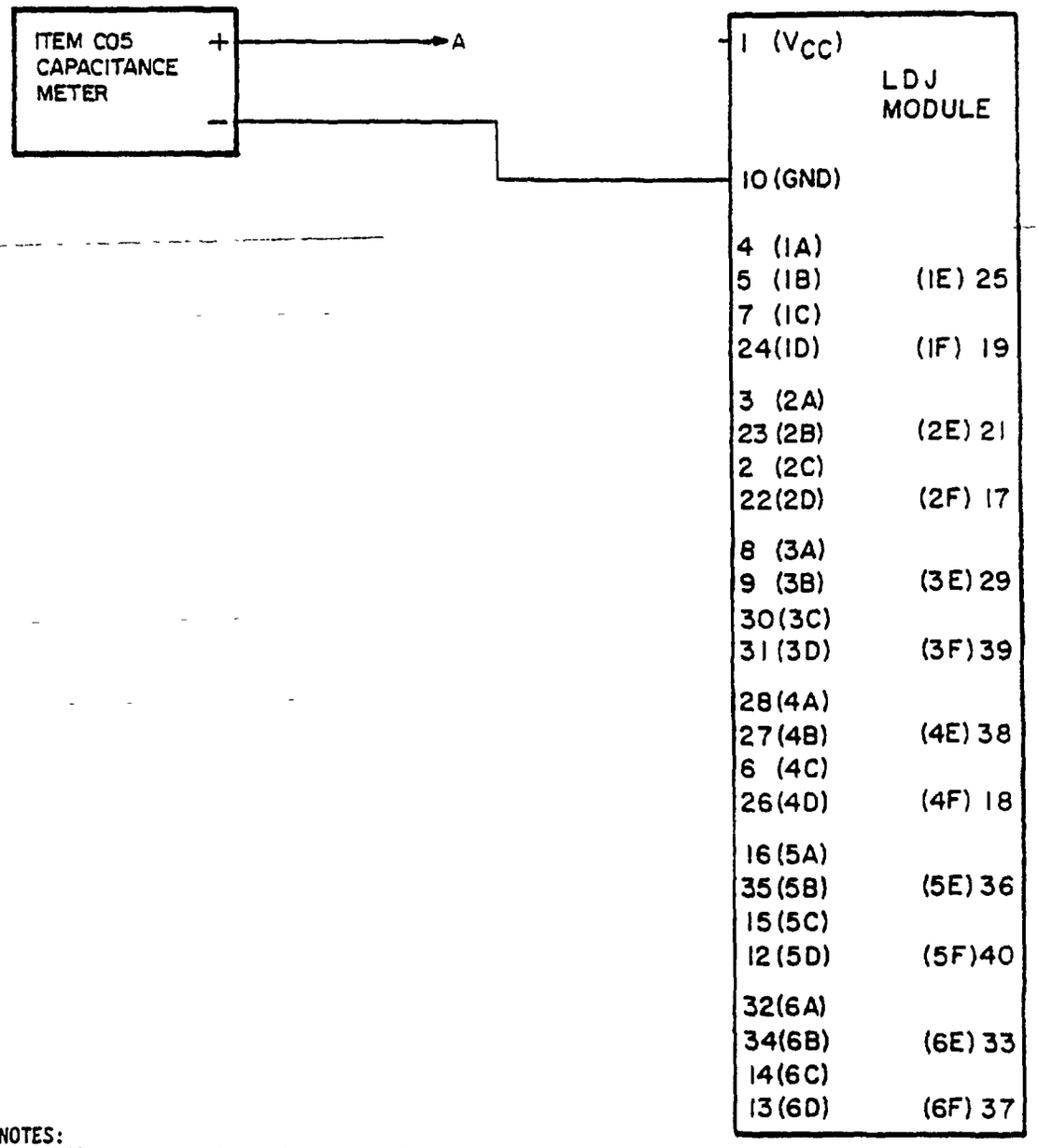
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
U1	1A Input capacitance	4	2	15	pF	17
U2	1B	5				
U3	1C	7				
U4	1D	24				
U5	2A	3				
U6	2B	23				
U7	2C	2				
U8	2D	22				
U9	3A	8				
U10	3B	9				
U11	3C	30				
U12	3D	31				
U13	4B	27				
U14	4A	28				
U15	4C	6				
U16	4D	26				
U17	5A	16				
U18	5B	35				
U19	5D Input capacitance	12	2	15	pF	17

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TABLE XIX. Input capacitance test -Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
U20	5C Input capacitance	15	2	15	pF	17
U21	6A	32				
U22	6B	34				
U23	6D	13				
U24	6C Input capacitance	14	2	15	pF	17

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 17. Input capacitance test circuit for table XIX.

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TABLE XX. Input clamp diode voltage test.

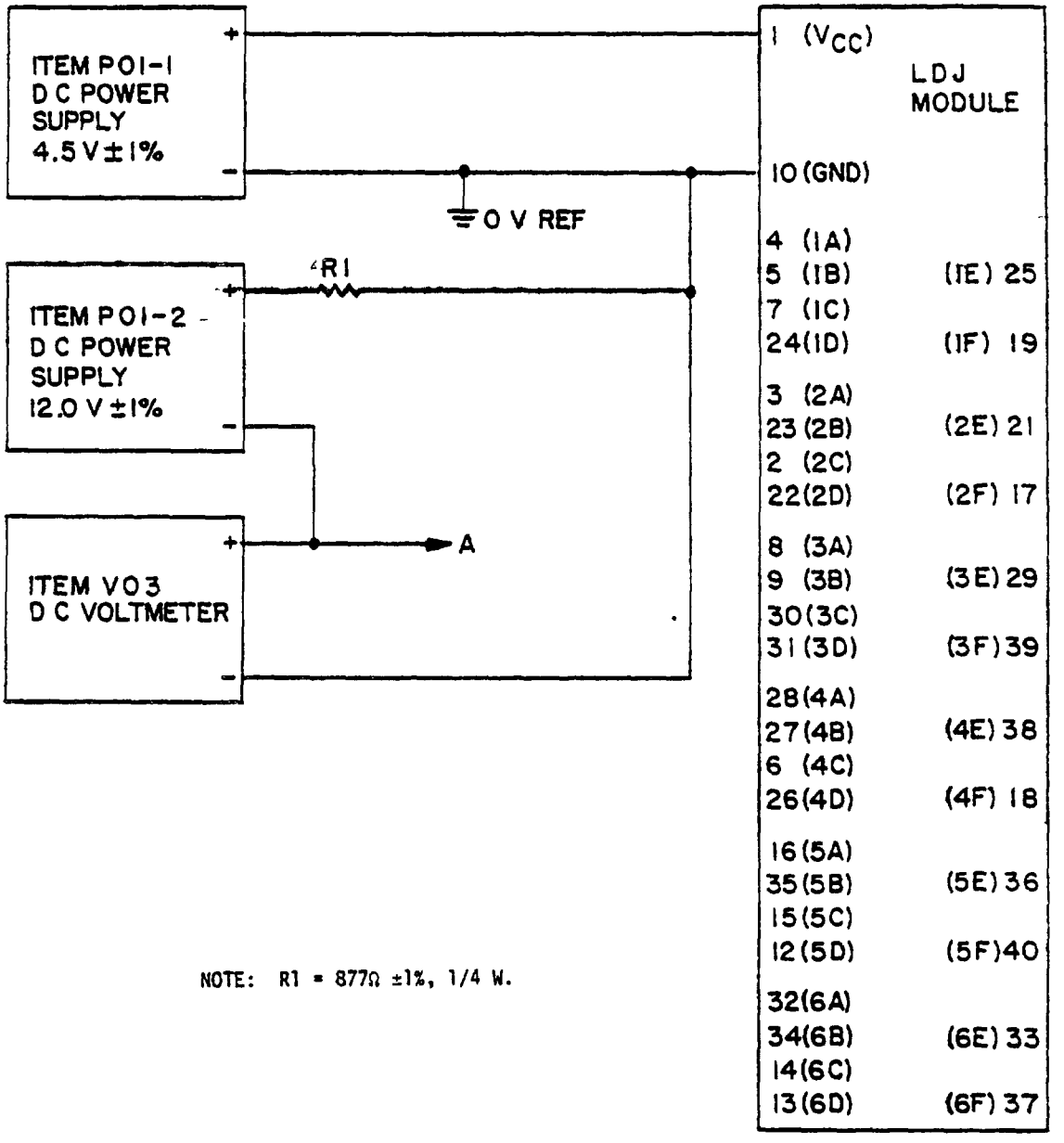
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
V1	1A Input clamp diode voltage	4	-0.4	-1.5	V	18
V2	1B	5				
V3	1C	7				
V4	1D	24				
V5	2A	3				
V6	2B	23				
V7	2C	2				
V8	2D	22				
V9	3A	8				
V10	3B	9				
V11	3C	30				
V12	3D	31				
V13	4A	28				
V14	4B	27				
V15	4C	6				
V16	4D	26				
V17	5A	16				
V18	5B	35				
V19	5C Input clamp diode voltage	15	-0.4	-1.5	V	18

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TABLE XX. Input clamp diode voltage test -Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
V20	5D Input clamp diode voltage	12	-0.4	-1.5	V	18
V21	6A	32				
V22	6B	34				
V23	6C	14				
V24	6D Input clamp diode voltage	13	-0.4	-1.5	V	18

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NOTE: R1 = 877Ω ±1%, 1/4 W.

FIGURE 18. Input clamp diode voltage test circuit for table XX.

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MIL-M- 28787/17A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE LDJ

This amendment forms a part of MIL-M-28787/17A(EC), dated 15 May 1979, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, first line: Delete "100°C" and substitute "85°C".

PAGE 2

1.2, Output current: Delete "14.4" and "16.0" and substitute "16.6" and "17.3" respectively.

PAGE 3

3.3, delete and substitute:

"3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.22 failures per million operating hours for class I or 0.57 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4)."

3.4, delete and substitute:

"3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4)."

PAGES 4, 5, AND 6

TABLE II: Delete and substitute table II as printed on pages 3 and 4 of this amendment.

PAGES 7 AND 9

4.3, delete and substitute:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGE 9

* 4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, B, G, T, and V are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, G, T, or V are part of a sequence of tests, the order of testing shall be as follows: T, A, B, G, V where applicable, followed by the remaining tests in the sequence."

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PAGE 9

* 4.5.1, lines four and five: Delete "C through P, S, and V" and substitute "C through F, H through P, and S".

* 4.5.3, line three: Delete "C through S and V" and substitute "C through F, H through S".

* 4.5.3, line five: Delete "C through S, U, and V" and substitute "C through F, H through S, and U".

PAGE 10

6.3, delete and substitute:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/17 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833411. Part number M28787/17-1 supersedes part numbers 2833411 and M28787/17."

Add new 6.4 as follows.

"6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4)."

PAGE 15

FIGURE 5: Delete and substitute new figure 5 as printed on page 5 of this amendment.

PAGE 21

* FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

PAGE 22

* FIGURE 8: Delete and substitute new figure 8 as printed on page 7 of this amendment.

PAGE 55

FIGURE 18: Delete and substitute new figure 18 as printed on page 8 of this amendment.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B24	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C18	Static output voltage, low level	V	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
D1-D18	Static output voltage, high level	VI	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
E1-E24	Input current, low level	VII	X	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	-0.40	-1.73	mA
F1-F24	Input current, high level	VIII	X	0	30	0	40	0	40	0	40	0	40	µA
G1-G12	Output short circuit current	IX	X	20	55	20	55	20	55	20	55	20	55	mA
H1	Power supply current, low level	X	S	1	54	1	54	1	54	1	54	1	54	mA
H2	Power supply current, high level	X	S	1	57	1	57	1	57	1	57	1	57	mA
J1-J12	Transition time, t_{TLH}	XI	S	4	45	4	50	4	50	4	50	4	50	ns
K1-K12	Transition time, t_{THL}	XI	S	4	20	4	25	4	25	4	25	4	25	ns
L1- L23 (Odd)	Propagation delay time, t_{PHL} ("E" outputs)	XII	S	4	19	4	22	4	22	4	22	4	22	ns
L2-L24 (Even)	Propagation delay time, t_{PLH} ("F" outputs)	XII	S	10	43	10	45	10	45	10	45	10	45	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M23 (Odd)	Propagation delay time, t_{PLH} ("E" outputs)	XIII	S	4	26	4	28	4	28	4	28	4	28	ns
M2-M24 (Even)	Propagation delay time, t_{PHL} ("F" outputs)	XIII	S	10	41	10	43	10	43	10	43	10	43	ns
N1-N11 (Odd)	Output crosstalk, low level	XIV	Q	0	560	0	560	0	560	0	560	0	560	mV
N2-N12 (Even)	Output crosstalk, low level	XIV	Q	0	800	0	800	0	800	0	800	0	800	mV
P1-P12	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
R	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient rise time	XVII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
T1	Isolation													
	Initial	XVIII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
T2,T3	Unused pin isolation													
	Initial	XVIII	Q	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	Q	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
U1-U24	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
V1-V24	Input clamp diode voltage	XX	Q	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V

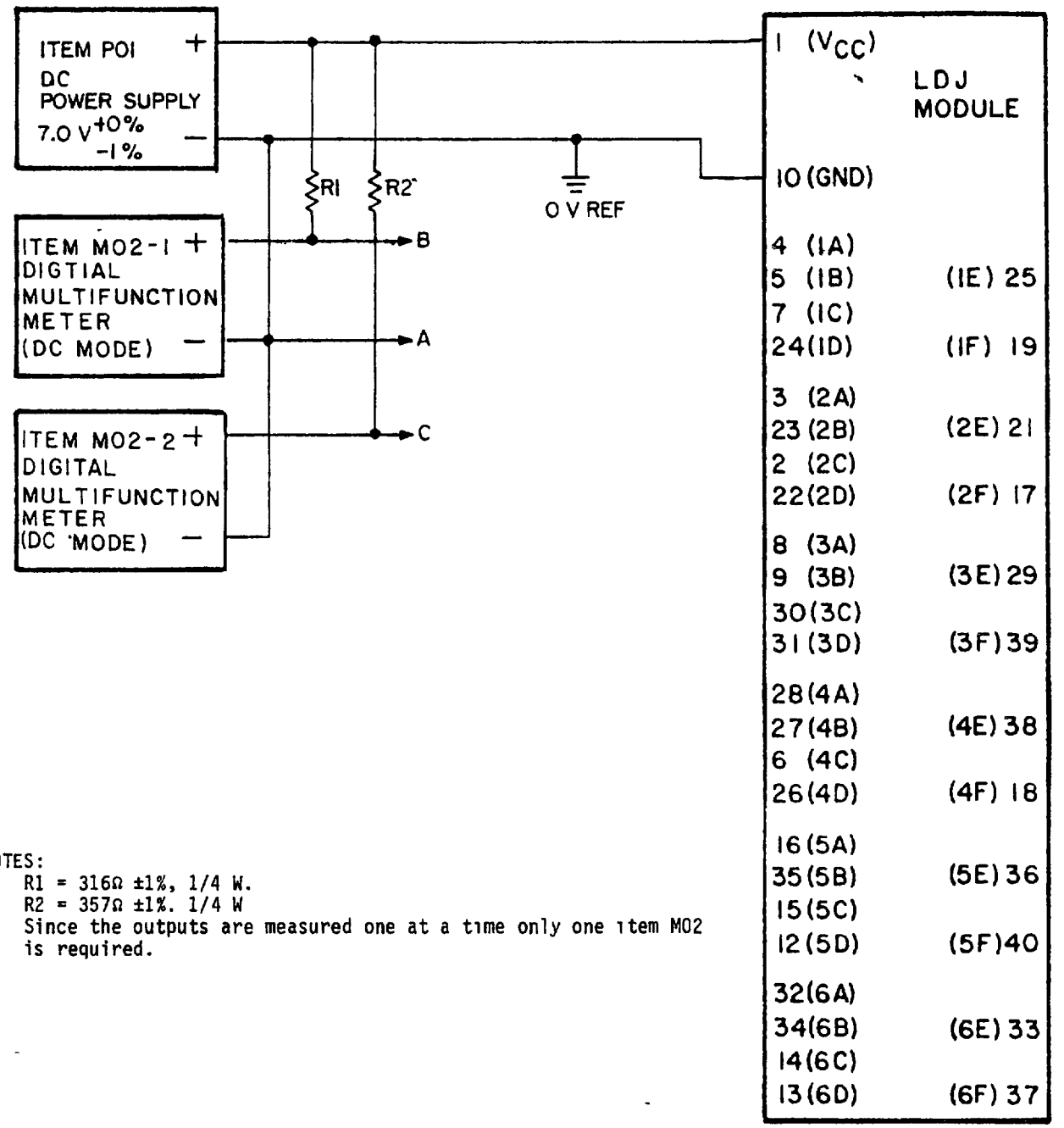
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

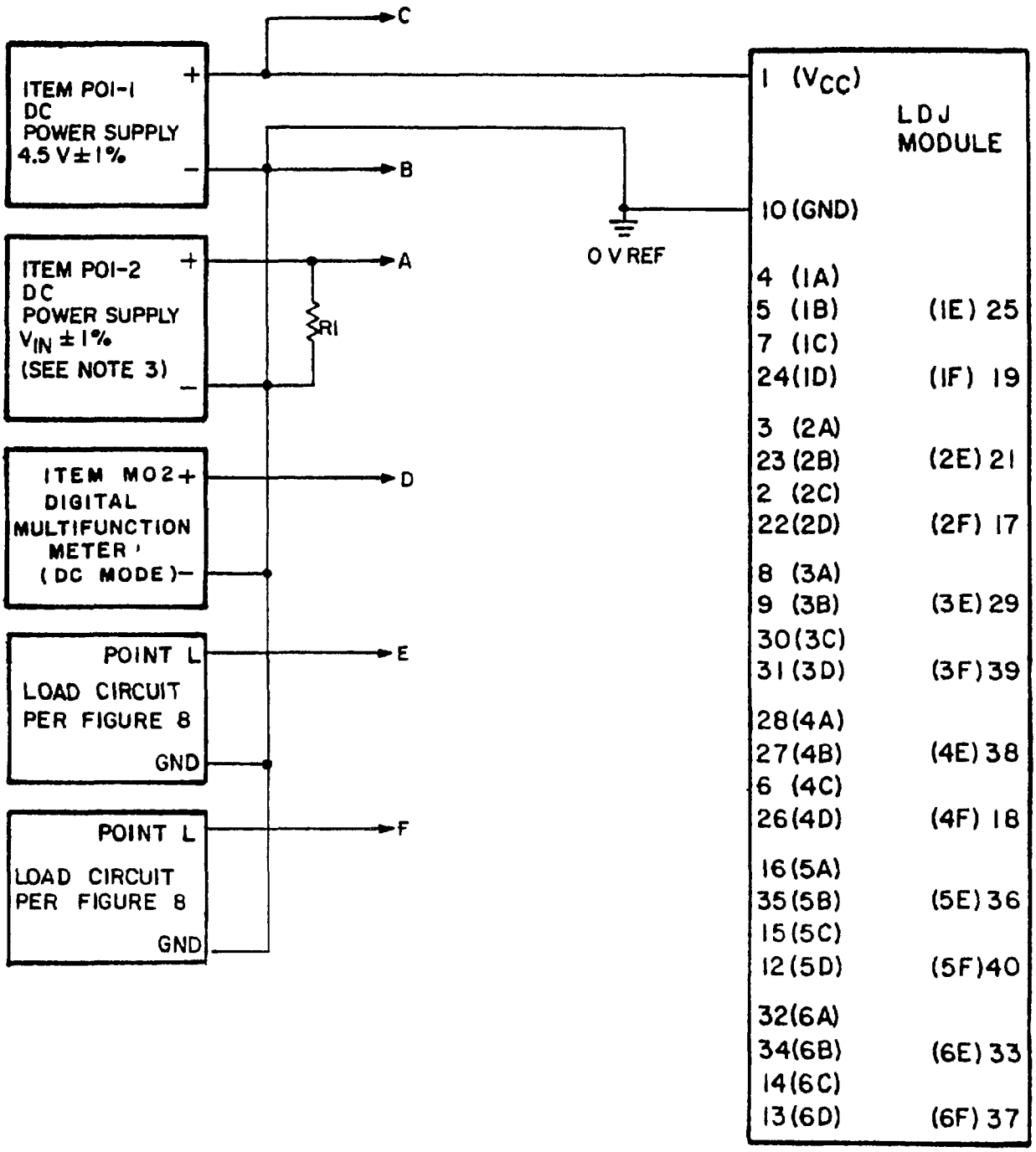
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NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. R2 = 357Ω ±1%. 1/4 W
3. Since the outputs are measured one at a time only one item M02 is required.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

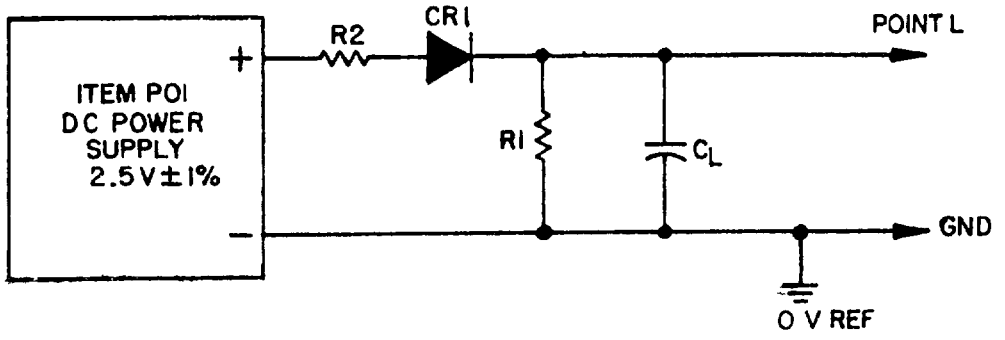
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NOTES:
1. R1 = 51Ω ±5% 1 W.
2. Both outputs of the circuit under test shall be loaded simultaneously.
3. V_{IN} = 2 V for test conditions C1-C12 and D13-D18. V_{IN} = 800 mV for test conditions C13-C18 and D1-D12.

FIGURE 7. Static output voltage (low-level and high-level) test circuit for tables V and VI.

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NOTES.
1. Load circuit 1 for outputs 1F-6F:

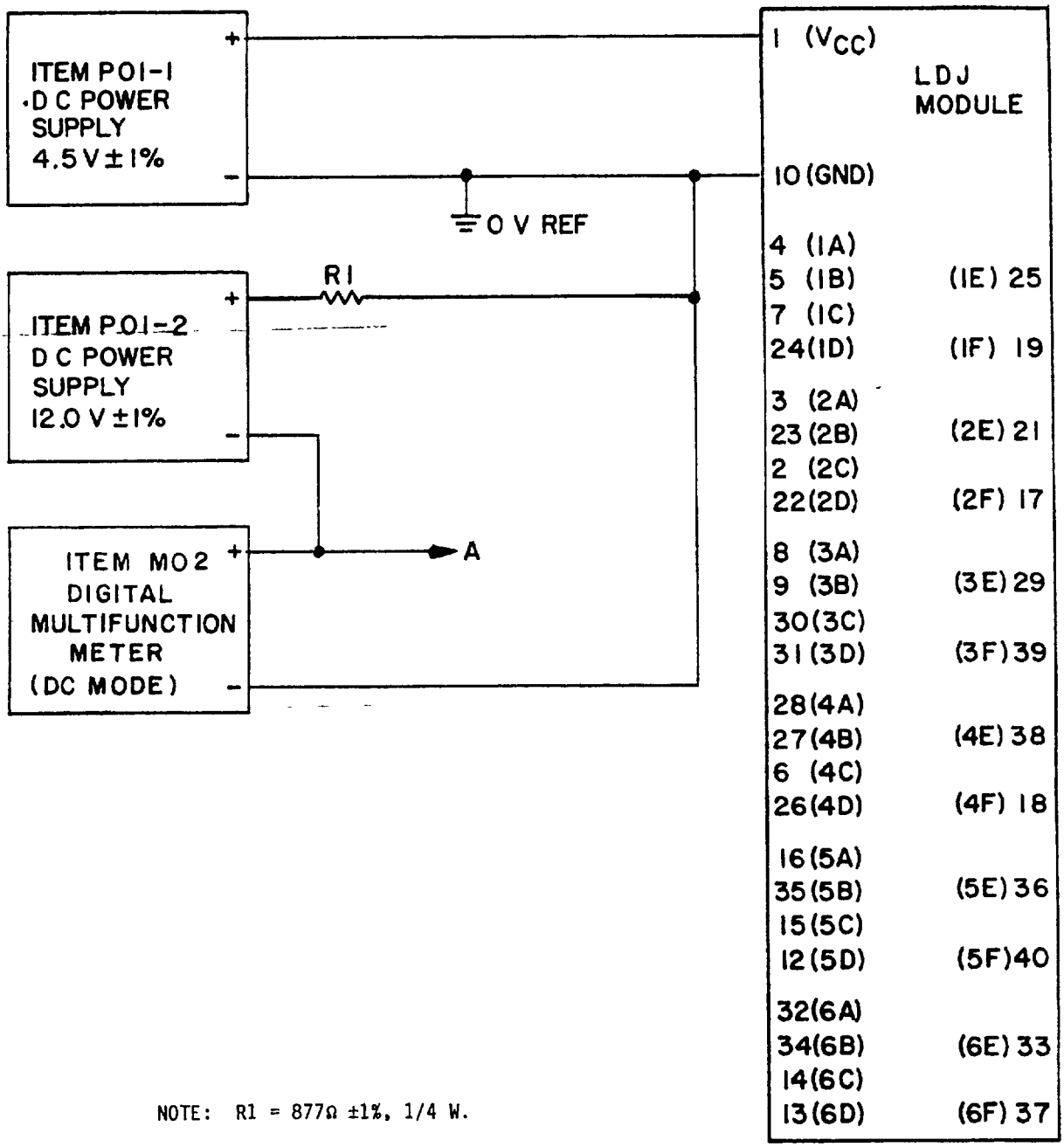
   a. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions J, K, L and M.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N and P.
   c. CR1 = 1N4150 or equivalent.
   d. R1 = 6.04 kΩ ±1%, 1/4 W.
   e. R2 = Select value for current flow of 16 mA ±5% out of point L with point L held at 400 mV.
   f. C_L is not required for dc electrical tests C1-C18 and D1-D18.
   g. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
   h. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

2. Load circuit 2 for outputs 1E-6E:

   a. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions J, K, L and M.
   b. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N and P.
   c. CR1 = 1N4150 or equivalent.
   d. R1 = 6.65 kΩ ±1%, 1/4 W.
   e. R2 = Select value for current flow of 14.4 mA ±5% out of point L with point L held at 400 mV.
   f. C_L is not required for dc electrical tests C1-C18 and D1-D18.
   g. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
   h. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 8. Logic load circuit.

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NOTE: R1 = 877Ω ±1%, 1/4 W.

FIGURE 18. Input clamp diode voltage test circuit for table XX.

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